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Fully 3D-Printed electric motor manufactured via multi-modal, multi-material extrusion

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ABSTRACT

Material extrusion additive manufacturing can process a wide variety of functional materials including electrically conductive, magnetic, and mechanically compliant polymer composites. While filaments developed for 3D printing often exhibit limited functionality, highly loaded functional composites originally formulated for specialised manufacturing processes can be processed via material extrusion. In this work, a commercial multi-material extrusion 3D printer was modified to process conductive inks, soft and hard magnetic composite pellets, and rigid and compliant polymeric filaments. Using this system, solenoids, hard magnets, and springs were fabricated. These components were combined through straightforward assembly to demonstrate the first fully 3D-printed electric motor — a linear actuator composed of five distinct functional materials: dielectric, electrically conductive, soft magnetic, hard magnetic, and flexible. The solenoids produced up to 2.03 mT magnetic fields, the magnets generated up to 71 mT magnetic fields, and the linear actuator attained a maximum displacement of 318 µm at its resonant frequency (41.6 Hz). This study demonstrates the capability of multi-modal, multi-material extrusion 3D printing to fabricate all critical components of electrical machines, with magnetisation of the hard magnets being the only post-printing step. This milestone advances multi-material, multi-functional 3D printing towards implementing in-situ, customised, low-waste, and low-cost functional hardware.

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1. Introduction

Electrical machines such as generators, transformers, and motors play a crucial role in the generation, distribution, and utilisation of electromagnetic energy. As energy demand increases and the shift toward clean, electricity-based solutions (e.g. electric vehicles) progresses, the ubiquity of electrical machines heightens interest in streamlining their fabrication to make them more cost-effective and accessible. Traditionally, the individual components of electrical machines (i.e. magnetic cores, wires, magnets, mechanical couplings), are manufactured separately through machining and assembly, before the complete device is put together. These operations involve specialised materials and equipment, often resulting in a labor-intensive and time-consuming process. Consequently, the development of a manufacturing technology capable of producing all the essential parts of electrical machines using a single piece of equipment and multipurpose feedstock would reshape the manufacturing landscape by enabling the single-step fabrication of these machines. Such a technology would profoundly impact disciplines such as robotics, space exploration, natural disaster relief, and education, among others, and it would advance the democratisation of electrical machine manufacturing, empowering institutions and individuals alike. To be viable, such a technology must be versatile in the geometries it can produce and capable of processing multiple functional materials. Ideally, it should also be cost-effective, low-waste, and easy to operate. These features align with the virtues of additive manufacturing (AM).

AM, or 3D printing, is a manufacturing technology that builds objects by adding material layer by layer. AM enables the fast and inexpensive fabrication of net-shape structures, facilitating the implementation of highly optimised shapes that are sometimes unattainable or prohibitively expensive to manufacture via traditional fabrication methods [1,2]. Additionally, AM

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enables the single-step fabrication of complex assemblies as integrated, highly precise structures, bypassing complex assembly operations and eliminating the associated risk of alignment errors [3,4]. Moreover, AM allows for low-waste, on-site fabrication of hardware, making it suitable for manufacturing in remote areas, including off-the-grid expeditions and space [5–7]. Furthermore, the accessibility and affordability of AM technologies have already empowered society by enabling small businesses and individuals to directly fabricate complex mechanical parts [8]. From do-it-yourself domestic gadgets to highly customised commercial products, AM has forever changed the way we approach prototyping and manufacturing small – and mid-batch mechanical parts [9]. In addition, recent demonstrations of 3D-printed devices exhibiting elastic [10–12], magnetic [13–15], optical [16], photonic [17], and electrical [18–20] properties attest to the potential of AM to reshape the manufacturing landscape of functional devices, including electronics and electrical machines [21–24].

Recent studies have demonstrated that AM technologies are capable of producing all the components involved in the fabrication of electrical machines, including magnetic cores [25–29], conductive coils [30–33], permanent magnets [13,34], mechanical couplings [35], and even control hardware [36]. Notably, Simpson and associates recently demonstrated modular, 3D-printed copper windings with an integrated heat exchanger intended for high power-density electrical machines [30], Lorenz and colleagues reported the AM of high-performance coils for switched reluctance electrical machines [37], Mettes et al. recently demonstrated the 3D printing of a stator for rotary electromagnetic motors [38], and Ellery and collaborators reported the construction of a DC motor combining additive and manual manufacturing [39]. These advances highlight the promise of AM for fabricating electrical machines. However, the reported implementations still rely on off-the-shelf parts (e.g. neodymium magnets, copper tape) to complete the devices. Thus, while the independent AM of electrical machine components has been demonstrated, their integrated fabrication using a single platform remains, to the best of our knowledge, unaccomplished. This challenge stems mainly from the inability of typical multi-material AM systems to process multiple high-performance materials simultaneously, as well as from the inherent difficulties in monolithically printing parts composed of materials with dissimilar processing requirements. We believe that the next critical step toward realising the full potential of AM for electrical machines is the development of a manufacturing system capable of producing all key components – conductive coils, magnetic cores, and permanent magnets – within a unified process. Such a technology would enable the fabrication of fully 3D-printed electrical machines and, ultimately, their single-step fabrication and integration into more complex 3D-printed devices.

Building on the previously outlined requirements for such a manufacturing technology – versatility, multi-material compatibility, cost-effectiveness, low-waste, ease of operation and accessibility – we propose multi-material extrusion as a promising technology for the fabrication of fully 3D-printed electrical machines. Multi-material 3D printing technologies have found applications in electronics [40], soft robotics [41], sports and mechanical metamaterials [42], and bone implants, scaffolds, and other biomedical devices [43–47], highlighting their versatility and motivating their use for fabricating fully 3D-printed electrical machines. Material extrusion is an AM technology that creates parts by dispensing fluid feedstock through a nozzle and depositing it layer by layer. It stands out among other AM technologies for its inherent suitability for multi-material fabrication [48,49], as well as for its cost-effectiveness and user-friendliness, which have made it one of the most widespread AM technologies [50,51]. While conventional material extrusion systems primarily process filament feedstocks, which exhibit limited functional performance [13], the technology is also compatible with alternative feedstock forms, such as inks [52] and pellets [53]. This versatility allows access to a broad range of high-performance materials, ultimately enabling the fabrication of more capable functional devices.

This study reports the development of a multi-modal, multi-material extrusion 3D printing platform capable of processing multiple highly doped, functional materials in filament, ink, and pellet form. Using this system, a linear motor composed of a solenoid, a planar spring, and permanent magnets is demonstrated. The first reported fully 3D-printed electric motor is a milestone that pushes the boundaries of 3D-printed functional devices and showcases the potential of AM for electrical machine fabrication. While it requires minor assembly, this motor serves as proof of concept of the production of fully 3D-printed electrical machines using a single, versatile platform and highlights the potential for their monolithic, single-step fabrication.

The remainder of this report is organised as follows: Section 2 describes the multi-modal, multi-material extrusion system developed in this work and presents the characterisation of the materials used in the fabrication of the 3D-printed electric motor. Section 3 describes each of the components that constitute the motor, detailing their design, fabrication challenges, and individual characterisation. Section 4 outlines the assembly

of the 3D-printed motor and its characterisation. Section 5 discusses the results and prospects of the proposed technology. Section 6 concludes the report.

2. Methods

This section describes the multi-modal, multi-material 3D printing system and the functional feedstock employed in this work. The devices 3D printed in this study were designed using the computer-aided design (CAD) software SolidWorks (Dassault Systèmes, Vélizy-Villacoublay, France), from which STL files were exported. The STL files were then arranged on the slicer software Simplify3D (Simplify3D, Cincinnati, OH, USA) and processed to produce G-code files that the 3D printer could interpret.

2.1. Multi-modal, multi-material extrusion 3D printer

The literature on electrically and magnetically functional devices 3D printed via material extrusion already includes magnetic cores [25–29], conductive coils [30–33], and permanent magnets [13,34], highlighting this technology's potential for fabricating electrical machines. Moreover, successful demonstrations of complex, multi-material systems, such as soft robotic grippers [54], vacuum pumps [10], and biostructures [55], further illustrate its versatility and open the door to fully 3D-printed, electromechanical devices. Additionally, material extrusion is a proven, effective technology for inexpensive and fast prototyping [56] and for the fabrication of highly customised devices [57]. These features make material extrusion an exceptionally promising technology for the fabrication of 3D-printed electrical machines, and for the democratisation of their manufacture. Multi-material extrusion is a variation of material extrusion that utilises multiple extruders sequentially, each dispensing a different material, enabling the monolithic integration of materials with complementary functional properties (e.g. dielectric and electrically conductive, rigid and flexible), which is critical for the fabrication of electrical machines.

The great majority of feedstock for material extrusion 3D printing is commercialised in the form of filament made of thermoplastics such as acrylonitrile butadiene styrene (ABS), polylactic acid (PLA), and nylon. These materials are apt for 3D printing due to their ability to melt, flow, and solidify in a controlled manner [58]. By mixing such thermoplastics with micro and nanostructured fillers, filament feedstock with a wide range of functional properties including electrical conductivity, magnetism, luminescence, and mechanical strength can be produced [59–61]. However, while the use of filament facilitates the printing operation (i.e. the material is fed at a constant, tightly controllable rate), the amount of functional filler that a material can contain before it becomes too brittle to be shaped as a filament is limited to about 55% in volume [13], constraining the functionality that can be embedded into the feedstock. Other feedstock formats, such as pellets (i.e. small nuggets of material that can be extruded with a screw setup) [53], ink [52], or paste [25,62], admit larger functional filler percentages. While the shape and rheology of such alternative feedstock make them more challenging to print than a filament, their higher filler content enables the fabrication of functional parts with better performance (e.g. higher electrical conductivity, higher magnetic permeability). By combining multiple forms of feedstock in a single printing operation, multi-modal, multi-material extrusion can leverage the properties of each feedstock format and enable the fabrication of highly optimised objects. Recent reports have demonstrated the implementation of such multi-modal, multi-material extrusion 3D printing machines. However, most works only integrate the hardware necessary to process two types of feedstock, and they overlook the potential of their combined use. For example, Martin et al. introduced a filament-pellet hybrid 3D printing system primarily focused on pellet extrusion for metal 3D printing, rather than integrating both filament and pellets in a single print [63]. Also, Engberg and colleagues described the installation of syringe pumps alongside filament extruders, but overlooked the potential of multi-modal 3D printing, focusing instead on the independent use of the syringe pumps for bioprinting [64]. In addition, Mettes et al. and Cañada and colleagues reported the implementation of hybrid filament-ink [32] and filament-pellet [33] 3D printers, respectively, for the fabrication of conductive coils, exploring the combined use of multiple feedstocks, but still limited to only two types.

In this work, a 3D printing system that simultaneously incorporates filament, pellets, and ink extruders was synthesised. The multi-modal, multi-material extrusion system is based on an E3D Motion System and ToolChanger (E3D – Online, Chalgrove, Oxfordshire, United Kingdom), which consists of a robotic handler and up to four different tools that can be picked up and used sequentially during a single print job. The original setup of the printer includes four filament extruders, each of which can be swapped for a different tool with a moderate amount of effort. In this work, three out of the four original filament extruders were swapped for a pellet extruder, an ink extruder, and a heater. The

tools that make up the final configuration of the machine are:

i. Filament extruder (Figure 1a): one of the original E3D Hemera direct drive filament extruders of the E3D Motion System and ToolChanger was kept in place. It features an E3D 24 V 30 W heater cartridge, an E3D thermistor cartridge, and a 0.4 mm nozzle.
ii. Pellet extruder (Figure 1b): a Mahor v4 70 W Pellet Extruder (Mahor.xyz, Spain) was incorporated to the system to enable 3D printing from pellets. A custom case was designed and 3D printed to adapt the pellet extruder to the E3D ToolChanger. The case wraps around the extruder and provides anchor points to the E3D toolhead plate and docking port, necessary to allow the pick-up and drop-off of the tool by the robotic arm.
iii. Ink extruder (Figure 1c): a syringe pump was custom-built from scratch, combining an E3D Hemera XS stepper motor, a lead screw, a linear rail, and custom-designed, 3D-printed parts, to enable 3D printing with inks. The syringe pump is designed to be compatible with the docking system of the E3D ToolChanger and accommodates a three-milliliter syringe that can be easily swapped, enabling seamless material exchange.
iv. Heater (Figure 1d): an E3D Hemera extruder with its nozzle and silicon insulation sock removed was installed to enable the curing of inks on the printer bed. During operation, the ink extruder and the heater can be used sequentially: first, the ink extruder prints a pattern; immediately afterward, the heater reproduces the printing trajectory of the syringe, drying the deposited ink as it hovers over it. This strategy enables the drying of ink while printing, facilitating the deposition of subsequent layers on top of the dried ink.

To accommodate the larger dimensions of the pellet and ink extruders compared to the original filament extruders, the docks that hold them to the printer frame while they are inactive were modified (Figure 2). Similarly, several parameters in the configuration files of the printer (e.g. tool offsets, type and number of

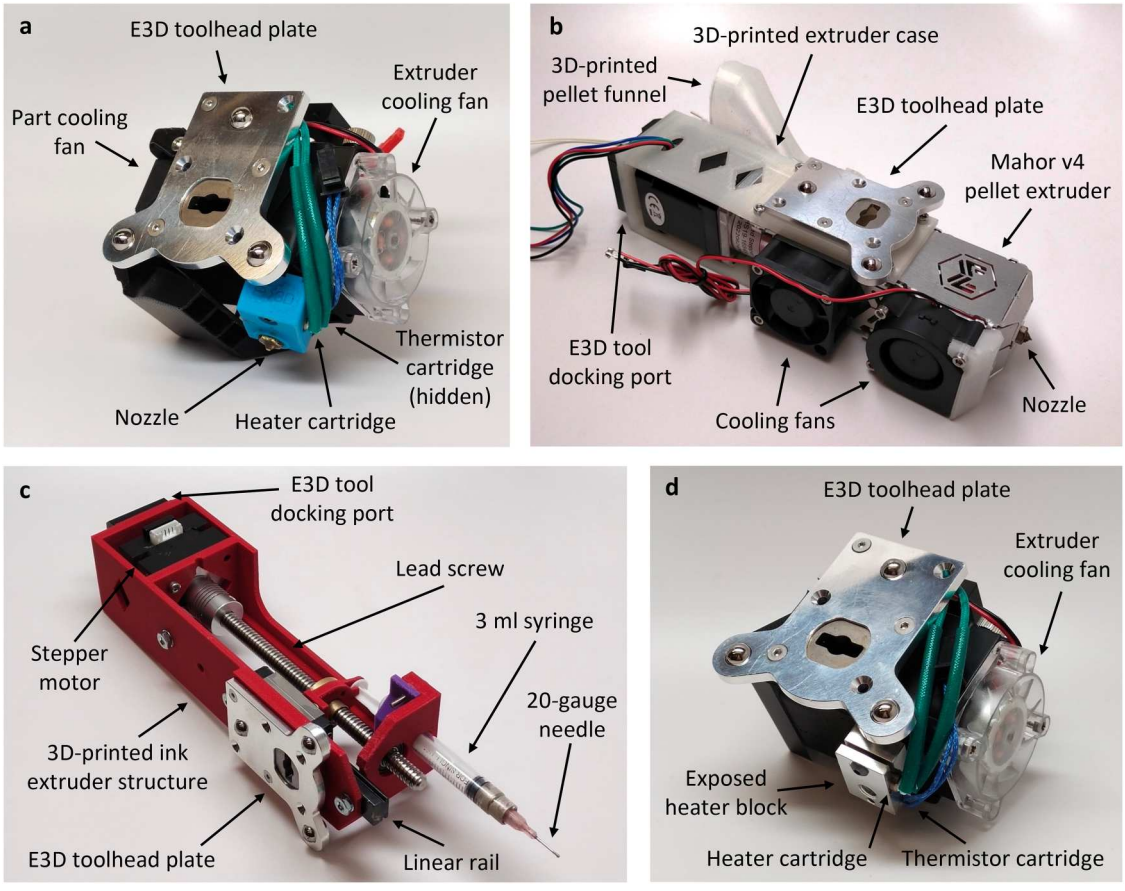


Figure 1. Tools integrated in the multi-modal, multi-material extrusion system developed in this work: E3D Hemera filament extruder (a), Mahor v4 pellet extruder with custom 3D-printed parts for E3D ToolChanger compatibility (b), custom-built ink extruder (c), and heater for ink curing (d).

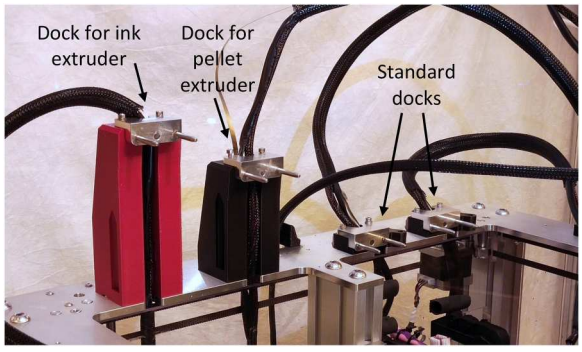


Figure 2. E3D ToolChanger docks adapted to the dimensions of the ink and pellet extruders.

thermistors and fans) were adjusted to the new tools. The original electronic hardware of the ToolChanger (Duet 2 Wifi and DueX5, Duet3D, Peterborough, United Kingdom) did not require adjustments to operate the new tools. The final configuration of the 3D printing system is shown in Figure 3. This multi-modal, multi-material extrusion 3D printer is capable of monolithically fabricating parts combining feedstock in the form of filament, pellets, and ink. The full system was built for under $4000 in parts.

2.2. Materials

The critical material properties for the fabrication of electrical machines are (i) electrical polarisation, (ii) electrical conductivity, and (iii) magnetism. Additionally, elasticity can introduce further functionality and versatility into 3D-printed electromagnetic devices. As previously explained, the ability to process feedstock in multiple forms enables access to a wide range of functional materials. In this section, various materials of interest for the AM of electrical machines are introduced and characterised.

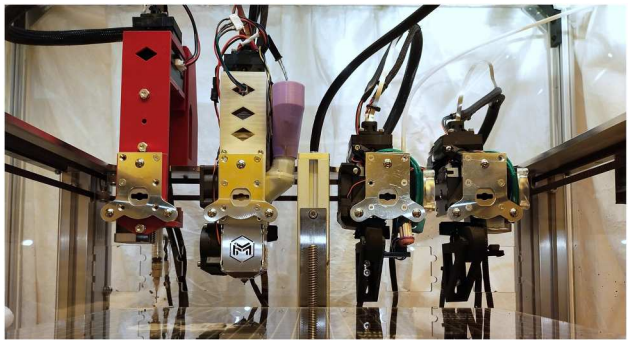


Figure 3. Lineup of tools in the final configuration of the proposed multi-modal, multi-material extrusion system. From left to right: ink extruder, pellet extruder, filament extruder, and heater.

2.2.1. Dielectric material

Most pure (i.e. undoped) feedstock for material extrusion 3D printing are dielectric polymers including ABS, PLA, nylon, polyethylene terephthalate (PET), polyether ether ketone (PEEK), and others. Each of these materials exhibits advantages and disadvantages regarding their printability, mechanical strength, thermal and chemical stability, toxicity, and so forth [50]. Among the available options, PLA stands out for its inexpensiveness, printability, non-toxicity, and environmental friendliness, while exhibiting modest mechanical strength and chemical stability [58,59]. Given that the devices fabricated in this work were not anticipated to be subject to high mechanical or chemical stresses, PLA was chosen as the base material for structural support and electrical insulation. Specifically, 3D-Fuel Pro PLA + (3DomFuel, Inc., Fargo, ND, USA) was selected as the dielectric feedstock. For the fabrication of advanced prototypes and final parts, materials with better heat and wear resistance, such as nylon, polycarbonate (PC) and polyether ketone ketone (PEKK) [50] should be considered, as well as composites with improved thermal conductivity [38].

2.2.2. Electrically conductive material

The potential of material extrusion 3D printing for the fabrication of electronics has led to the development and commercialisation of several electrically conductive filaments, often made of polymer-based composites doped with metallic (e.g. copper [20]) and carbon-based (e.g. graphene [65]) fillers. Among the commercially available electrically conductive filaments, Electrifi (Multi3D, Middlesex, NC, USA) – a copper-reinforced PLA filament – is, to the best of our knowledge, the most electrically conductive. However, its resistivity, in the order of $10^{-4}\,\Omega\cdot$m [20], is still considerably high – about four orders of magnitude larger than that of bulk copper. This circumstance suggests the need to consider other forms of printable feedstock, such as inks and pastes, to be able to attain higher conductivities.

Conductive inks and pastes typically consist of highly conductive particles suspended in a fluid medium that can contain additives to prevent issues such as clogging and doping agglomeration. Multiple electrically conductive inks and pastes have been developed using silver, copper, gold, and carbon as fillers [66,67]. Recent advances in direct ink writing (DIW) – a variation of material extrusion that uses syringes to dispense inks and pastes – reveal great potential for the fabrication of electronic components using inks [68,69]. However, most conductive inks commercially available are intended for two-dimensional printing (e.g. screen

printing, roller-based techniques), posing challenges for their adoption for material extrusion 3D printing. Additionally, post-printing drying or curing treatments, such as high-temperature annealing or UV light exposure, are usually required to achieve solidification and reach high electrical conductivity. However, these treatments are often incompatible with polymer-based 3D printing because typical curing temperatures exceed the melting point of many polymers, and UV light is known to accelerate the degradation of PLA [70]. These circumstances limit the range of inks and pastes suitable for material extrusion, highlighting two main selection criteria: (i) adequate rheology: the material must flow smoothly through a syringe without clogging it, while being sufficiently viscous to prevent its spreading beyond the intended deposition area; and (ii) compatible drying process: the material must reach its solid state without the need for excessive heat or UV light. Following these criteria, PriElex AG-1074 Silver Conductive Ink (Heraeus Precious Metals North America, West Conshohocken, PA, USA), a low-temperature-curing (80°C) silver ink, was selected for the fabrication of conductive traces in this work. Figure 4 shows a conductive trace of PriElex AG-1074 3D printed on a PLA substrate using the multi-modal, multi-material extrusion system previously described.

These silver ink traces, which were also dried on the same machine using the heater described in Section 2.1, exhibit a resistivity of $\sim 7 \times 10^{-7}$ Ω·m – three orders of magnitude lower than that of Electrifi. Compared to copper-doped PLA, the low resistivity of silver ink enables (i) the supply of larger currents to 3D-printed devices without risking excessive heat dissipation and (ii) the implementation of smaller conductive traces while keeping the device resistance low, facilitating the fabrication of more compact devices. Traces as thin as 30 µm in height were reliably 3D printed with PriElex AG-1074 (Figure 5).

2.2.3. Soft magnetic material

The performance of electrical machines is closely tied to that of their magnetic cores, which are made of soft magnetic materials, i.e. materials that easily magnetise and demagnetise, enabling the guiding and intensification

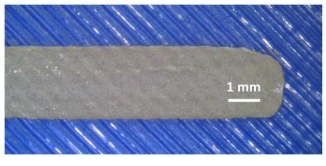


Figure 4. Trace of PriElex AG-1074 silver ink 3D printed on a PLA substrate.

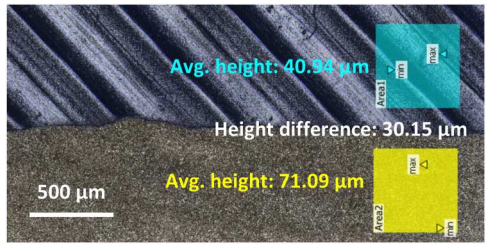


Figure 5. Laser scanning confocal microscope (Keyence VK-X, Keyence, Itasca, IL, USA) image and height measurements of a trace of PriElex AG-1074 silver ink 3D printed on a PLA substrate. The height of the silver ink trace is the difference between the average height of each feature.

of external magnetic fields. Multiple studies have reported the development of AM-compatible soft magnetic feedstock [71] including filaments, pastes, and pellets. Soft magnetic feedstock in the form of composite, thermoplastic-based filaments have been shown to perform poorly, reaching relative magnetic permeabilities below 10 [27,28,72–76]. This is attributed to the inability to fabricate filaments suitable for 3D printing with high infill content. Meanwhile, soft magnetic printable feedstock in the form of paste has been shown to yield relative magnetic permeabilities of several hundred [77,78]. However, such materials require high-temperature (950°C or higher) post-processing, which is incompatible with the thermoplastics-based, multi-material 3D-printing framework explored in this study. Alternatively, recent work has shown that soft magnetic feedstock in the form of pellets can yield a competitive relative magnetic permeability, in the order of 30, without requiring any post-processing [29]. This work utilises pellets of nylon 12 doped with anisotropic FeSiAl powder at 50% in volume, 86% in weight (MATE Co., Okayama, Japan) – the same material characterised in [29] – to enhance the magnetic performance of silver ink-based, 3D-printed solenoids. The soft magnetic material does not exhibit bulk electrical conductivity, eliminating the concern of eddy currents affecting the performance of the fabricated devices. The FeSiAl-doped nylon pellets can be processed using the reported multi-modal, multi-material extrusion system, using the pellet extruder described in Section 2.1.

2.2.3.1. Hard magnetic material. Unlike soft magnetic materials, hard magnetic materials retain a remanent magnetisation in the absence of an applied magnetic field, enabling their use as permanent magnets in applications such as energy harvesting, actuation, sensing, and electromagnetic structure anchoring. Like in the case of soft magnetic cores, the 3D printing of hard magnets has been attempted using feedstock in the

form of filament [13,79–81] and paste [82], facing similar challenges: filament feedstock admits limited functional filler content, constraining their performance; also, pastes require high-temperature sintering to secure their shape and properties, which is incompatible with polymer-based multi-material manufacturing. The AM of permanent magnets from pelletised feedstock has been attempted via Big Area Additive Manufacturing (BAAM) [83–85], a material extrusion technology that stands out for its compatibility with highly-doped materials and ability to produce large parts, but exhibits low resolution due to its large nozzle size (e.g. 7.6 mm [84], 5.08 mm [85]). In this study, a pellet-based composite made of nylon 12 and particles of strontium ferrite at 69% in volume (MATE Co., Okayama, Japan) was used for the fabrication of hard magnetic components. Like with soft magnetic materials, printing hard magnets directly from pellets enables access to high functional filler contents, while circumventing the need for high temperature post-processing. The strontium ferrite-doped nylon pellets used in this study can be processed using the pellet extruder installed in the multi-modal, multi-material extrusion 3D printer introduced in Section 2.1, equipped with a 0.4 mm-diameter nozzle. The fabricated components need to be magnetised after printing using an external magnetic field to become hard magnets.

To characterise the magnetic hysteresis of the strontium ferrite-doped nylon composite, a cylindrical disk of volume 0.0083 cm^3 was 3D-printed and placed between the poles of a vibrating sample magnetometer (VSM; Model 1660, ADE Technologies, Westwood, USA). The field generated by the VSM was swept from −1 T to +1 T to obtain the full B-H loop (Figure 6). Because the measured hysteresis curve includes the contribution of the sample's demagnetising field, a correction was applied to obtain the intrinsic curve, which reflects only the material response independent of geometry. The demagnetising factor N for the printed specimen was estimated from its aspect ratio to be $N \approx 0.07$. Fitting the linear slope of the measured high-field region yielded $N = 0.0733$, in close agreement with the theoretical value and validating the correction procedure. Using the intrinsic curve, the remanent induction and intrinsic coercivity of the hard magnetic composite were found to be $B_r = 0.1364\ T$ and $H_{ci} = 0.289\ T$, respectively, and the corresponding remanent magnetisation was calculated to be $M_r = 1.09\ x10^5\ A/m$.

2.2.4. Flexible material

While electrically and magnetically functional materials are the critical integrators of electrical machines, materials with other properties are required to implement a wide variety of devices. In particular, the use of flexible

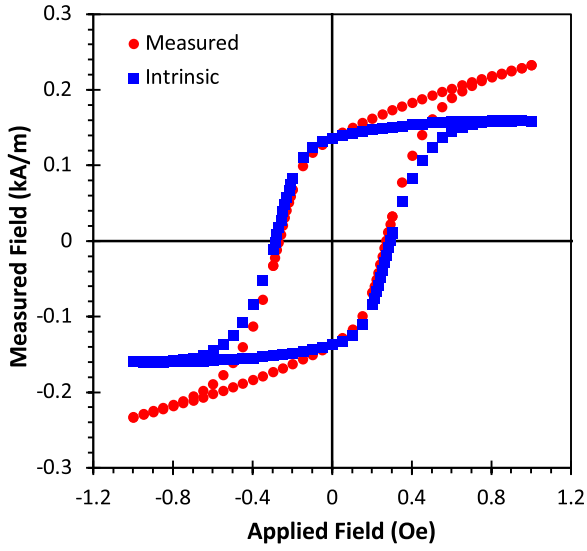


Figure 6. Magnetic hysteresis curve (i.e. B-H curve) of the strontium ferrite-doped nylon hard magnetic composite used in this work.

materials enables the fabrication of pressure [86] and strain [87] sensors, membrane actuators [13], pumps [10,12], and other transducers [88], which holds promise for the development of flexible electronics and soft robotics. Flexible materials for material extrusion are commercially available in the form of filament, e.g. NinjaFlex thermoplastic polyurethane (TPU) filament (NinjaTek, Manheim, PA, USA) and FiberFlex 40D (Fiberlogy, Brzezie, Poland). Printing with such filaments requires the use of direct drive filament extruders, in which the motor that pushes the filament through the nozzle sits on the extruder head, minimising the distance that the filament travels between the driving gear and the nozzle. In this work, FiberFlex 40D, a TPU filament with a Shore hardness of 40D, was used to fabricate compliant features. The material consists of 96% thermoplastic copolyester elastomer and 4% additives, according to the technical data sheet provided by the manufacturer. Lee et al. modelled the mechanical behaviour of FiberFlex 40D under large elastic strains using a five-parameter Mooney-Rivlin hyperelastic model, previously fitted to experimental tensile data. The parameters obtained were $C_{01} = 18.1$ MPa, $C_{02} = -2.66$ MPa, $C_{10} = -9.49$ MPa, $C_{11} = 2.61$ MPa, and $C_{20} = 0.2$ MPa, with a density of 1.16 g/cm^3 [10].

3. Fabrication and characterisation of fully 3D-printed electric motor components

The critical elements that make up most electrical machines are electrically conductive solenoids, soft magnetic cores, permanent magnets, and mechanical

couplings. In this study, all these components were fully 3D printed using the multi-modal, multi-material extrusion machine and materials described in Section 2. The components were experimentally characterised, with a sample size of one for both 3D-printed magnets and solenoids.

3.1. Fully 3D-printed, cored solenoids

The literature on 3D-printed solenoids manufactured via multi-material extrusion includes reports of three-dimensional, multi-layer, monolithically-fabricated, air-cored [32,89] and soft magnetic-cored [33,90] solenoids, demonstrating the potential of material extrusion for the fabrication of electromagnets. Recently, Mettes et al. and Cañada et al. independently advanced the state of the art of 3D-printed solenoids by introducing the use of highly conductive silver ink [32] and high-permeability soft magnetic pellets [33], respectively. This study integrates both breakthroughs using the multi-modal, multi-material extrusion system described in Section 2, which enables the simultaneous leveraging of multiple high-performance functional materials. An AM-friendly solenoid design consisting of stacked conductive Archimedean spirals of alternating spin, separated by thin dielectric layers, was chosen for the implementation of the coils. The nominal dimensions of the solenoids are summarised in Table 1. The footprint of the solenoids was confined to a 25 mm-diameter circle to keep the design compact and facilitate a fair comparison with 3D-printed solenoids reported in the literature. Figure 7 summarises the workflow to monolithically 3D printing the high-performance cored solenoids reported in this study, which is described in detail below.

The dielectric features of the solenoids (i.e. structural support and insulation layers) were printed in 3D-Fuel Pro PLA + using the default filament extruder of the E3D ToolChanger 3D printer. The conductive paths were printed in PriElex AG-1074 Silver Conductive Ink using the custom syringe pump described in Section 2, and the soft magnetic cores were fabricated in FeSiAl-doped nylon using the pellet extruder also introduced in Section 2. The printing settings employed in each case, optimised experimentally via iteration, taking the feedstock's manufacturers' recommendations into account, are summarised in Table 2. The bed temperature was determined considering only the material that was printed directly on the printer bed – in this case, pure PLA – and kept consistent across printing settings to maintain a stable bed temperature during the whole printing operation. In addition to typically relevant parameters, Simplify3D's *retraction distance* – which determines how far back the feedstock in an extruder is retracted between subsequent depositions within the same layer to avoid oozing – and *tool change retraction distance* – which controls how far back feedstock is retracted when switching from one tool to another to prevent material from dripping from inactive tools – were found to be critical to attaining adequate multi-modal operation, since different feedstocks exhibit different rheology. In particular, it was determined experimentally that the pellet extruder operates better with a reduced retraction distance, and that the ink extruder should barely retract (see Table 2). Moreover, an inconsistency in the 3D printing instructions generated by Simplify3D (i.e. the gcode) motivated the addition of an auxiliary feature to the process, labelled *Aux* in Table 2: it was observed that, after a tool change, the generated gcode forced the new active tool (tool A) to push out as much feedstock as the recently deactivated tool (tool B) had retracted, instead of the amount that tool A had retracted the previous time it had been used. This mismatch resulted in the following scenario: PLA would be printed out of the filament extruder and, before swapping tools, the filament would be retracted 10 mm inside the extruder; afterward, the ink extruder would become active and push out the ink equivalent to 10 mm of filament, instead of the desired 0.01 mm, flooding the sample. This was resolved in this work by creating the Aux feature and using it as buffer between features that require different tool change retraction distances: after using the filament extruder and retracting the filament 10 mm, the heater (i.e. a tool with no feedstock) became active and virtually extruded the 10 mm imposed by the tool change, to then reset the value to the desired 0.01 mm before switching to the ink extruder. Additionally, to ensure proper flow of material after a tool change, dummy structures were systematically 3D printed: after every tool change, and prior to printing features on the main body of the solenoid with the newly active extruder, a dummy structure was printed on the side.

Table 1. Summary of the nominal geometrical dimensions of 3D-printed solenoids.

Parameter name	Value
Conductive trace height	25 µm
Conductive trace width	1 mm[a]
In-plane conductive loop centre-to-centre radial distance (pitch)	2.8 mm
Inner diameter of conductive spiral	5 mm
Outer diameter of conductive spiral	<25 mm
Number of turns	3
Out-of-plane insulation thickness (inter-spiral distance)	150 µm
Soft magnetic core diameter	3 mm

[a]Even though the conductive trace was designed to be 1mm-thick, it was printed thicker because of the zigzagging motion used to guarantee continuous ink deposition, detailed below.

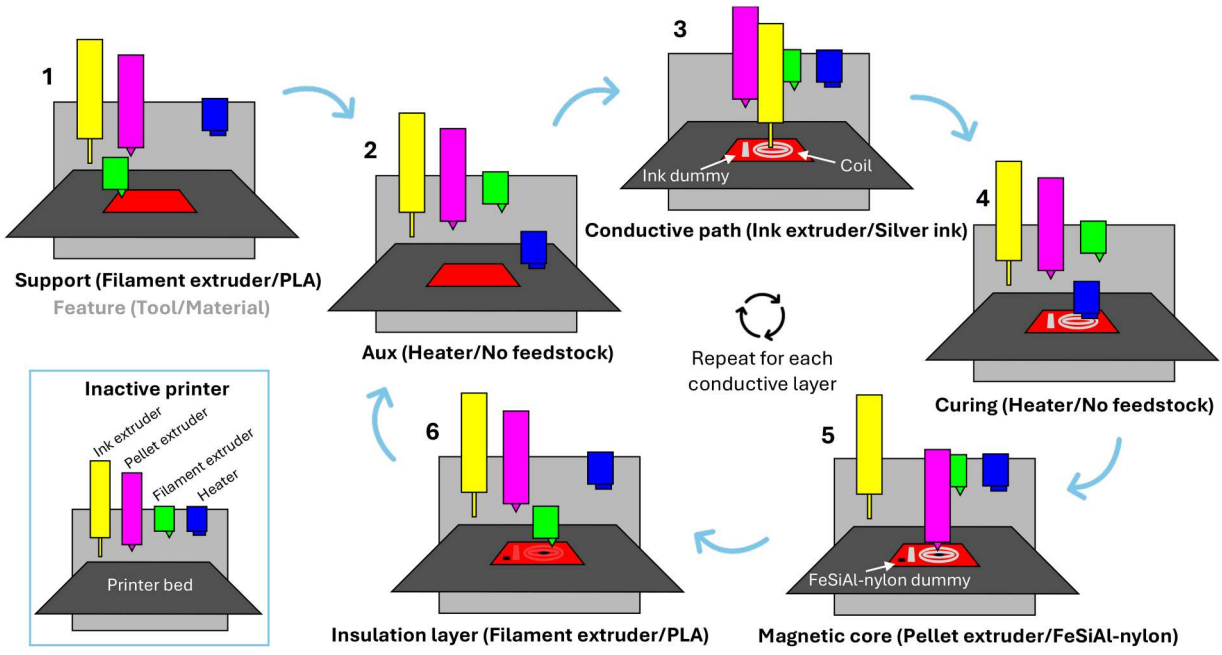


Figure 7. Workflow schematic for 3D printing monolithic, FeSiAl-nylon-cored, silver ink-based solenoids.

A 20-gauge blunt needle was used to dispense the silver ink, as it was found that 22-gauge needles were prone to clogging, and 18-gauge needles would drip while at rest. Preliminary silver ink printing tests yielded discontinuous traces when depositing the silver ink with a single stroke of the needle. To avoid this, the extrusion width parameter was tuned experimentally to induce a zigzagging motion on the needle that causes it to behave as both an ink dispenser and a brush. The extrusion width parameter determines the width of the trace that a tool is meant to deposit with each stroke, and it is typically set to a value comparable to the diameter of the nozzle. An extrusion width significantly smaller than the width of a feature to be printed results in the generation of an infill path (i.e. multiple strokes of the tool are used to sweep the full surface of the feature). Additionally, if the value assigned to the extrusion width is smaller than the actual nozzle diameter, the slicer software generates a tool path with narrow strokes that result in overlapping sweeps of the nozzle. In this case, such conditions were artificially imposed by assigning a very small value to the extrusion width parameter (0.05 mm versus the ~1 mm of both the conductive trace width and the needle diameter), prompting the slicer software to generate a fine zigzagging path that resulted in overlapping sweeps of the needle tip (Figure 8). Consequently, the needle effectively behaves as both an ink dispenser and a brush, compensating for possible inconsistencies in ink flow and guaranteeing the continuity of the conductive path.

Once the silver ink is deposited, the heater tool hovers over it to dry it, reproducing the same pattern in a single, slow stroke. The temperature of the heater, its hovering distance over the printed part, and its movement speed were optimised experimentally to enable the thorough curing of the silver ink traces while preventing the melting of the PLA structures already present in the printed object. An infrared camera (B20, HIKMICRO, Zhejiang, China) was used during curing tests to monitor the temperature at the surface of the printed part, targeting ~80°C – the curing temperature listed by the manufacturer of the silver ink. A heater temperature of 300°C, a vertical distance of 1.5 mm from the printed sample, and a printing speed of 30 mm/min were found to yield adequate annealing of the silver ink imprints without compromising the PLA structure.

Using the described process, air-cored and FeSiAl-nylon-cored, silver ink-based solenoids were monolithically 3D printed (Figure 9). Additionally, a two-step fabrication process for FeSiAl-nylon-cored solenoids was implemented: a Ø 2 mm rod was extruded from FeSiAl-nylon pellets using a Filabot EX2 filament extruder (Filabot Corporation, Barre, VT, USA). The rod was then cut to the appropriate length and inserted into the centre of a 3D-printed air-cored solenoid. This approach was expected to yield denser cores, as it was identified through observation that the fully monolithic fabrication process yielded porous, geometrically irregular cores (Figure 10).

The magnetic field generated by the solenoids was characterised and compared with the literature. An experimental apparatus consisting of a gaussmeter

Table 2. Summary of key printing parameters used to 3D print FeSiAl-nylon-cored, silver ink-based solenoids.

Feature	Tool	Material	Nozzle diameter	Layer height (µm)	Extrusion multiplier (-)	Extrusion width (mm)	Nozzle Temp. (°C)	Bed Temp. (°C)	Speed (mm/min)	Retraction distance (mm)	Tool change retraction distance (mm)
Support	Filament extruder	Pure PLA	0.4 mm	200	0.9	0.48	200	70	6000	4	10
Insulation layers	Filament extruder	Pure PLA	0.4 mm	50	1.2	0.48	200	70	2000	4	10
Conductive path	Ink extruder	Silver ink	20-gauge	25	1.2	0.05	-	70	1000	0	0.01
Curing	Heater	No feedstock	-	-	-	0.7	300	70	30	-	10
Magnetic core	Pellet extruder	FeSiAl-nylon	0.8 mm	200	1.0	0.8	270	70	1000	1	10
Aux	Heater	No feedstock	-	-	-	-	300	70	-	-	0.01

equipped with an axial probe (HGM0900, MAGSYS magnet systeme GmbH, Dortmud, Germany), an Agilent E3644A DC power supply (Keysight Technologies, Santa Rosa, CA, USA), a non-magnetic positioning stage (Altos Photonics, Inc., Bozeman, MT, USA), a Mu-metal magnetic shielding chamber (Magnetic Shield Corporation, Bensenville, IL, USA), and a custom 3D-printed support structure fabricated in undoped PLA was used to extract high resolution (10 mG measurement error – from the manufacturer of the gaussmeter) measurements of the magnetic flux densities generated by each printed solenoid for a range of applied currents at the intersection between its axis and its top layer. Figure 11 shows the magnetic flux densities generated by four-layered solenoids featuring an air core, a monolithically fabricated FeSiAl-nylon core, and an inserted FeSiAl-nylon core. The data corresponding to the solenoids with an air core and an inserted FeSiAl-nylon core were obtained using the same 3D-printed solenoid, with and without and inserted core; the data for the monolithically fabricated FeSiAl-nylon core was obtained from a separate, geometrically identical solenoid sample fabricated together with its core in a single 3D printing step. The data show that the monolithic integration of the soft magnetic core only achieves a modest enhancement of about 4% compared to the air-cored implementation, while the inserted FeSiAl core yields a notable increase of about 68% in the magnetic field that is generated. This difference in performance is attributed to the higher density of the independently extruded core. Moreover, the monolithic implementation features a wider core (Figure 10), which facilitates 3D printing, but reduces the ability of the core to concentrate magnetic flux lines at a specific location. The performance gap between the monolithically 3D-printed FeSiAl-nylon core and the inserted one suggests that the monolithic process requires improvement to adequately leverage the potential of the soft magnetic core. A possible strategy is the substitution of the 0.8 mm nozzle incorporated on the pellet extruder for a 2 mm one, enabling the vertical extrusion of a Ø 2 mm rod into the central cavity of a solenoid after the rest of the features have been printed, emulating the two-step fabrication.

The magnetic flux density generated by eight-layered solenoids with an air core and an inserted FeSiAl-nylon core is compared in Figure 12 to the values reported in the literature for other fully 3D-printed solenoids of similar geometries (i.e. eight stacked layers, active area of diameter under 30 mm). The values obtained from the literature correspond to solenoids that use copper-doped PLA as conductive material [33,89,90], instead of silver ink. To the best of our knowledge, no magnetic field data has previously been reported form silver ink-

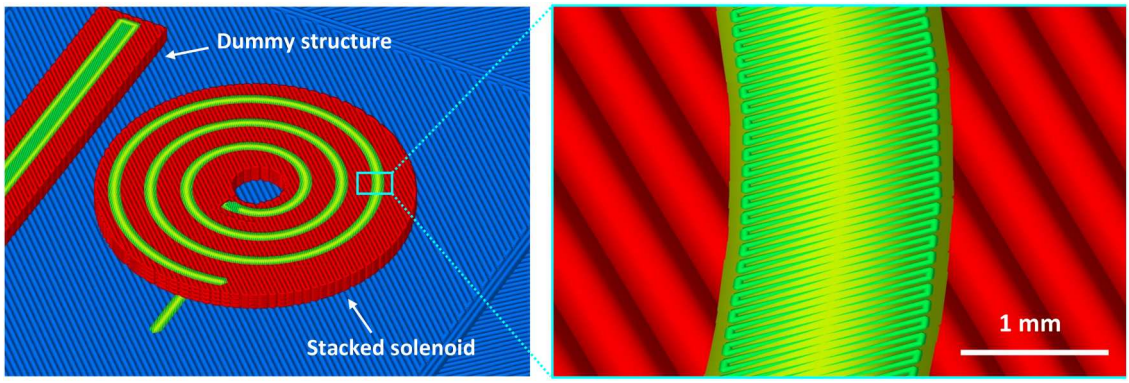


Figure 8. Detail of the 3D-printing path followed by the extruders during silver ink printing. Each feature is identified by a colour: conductive path (green), insulation layers (red), support (blue), curing trajectory (yellow).

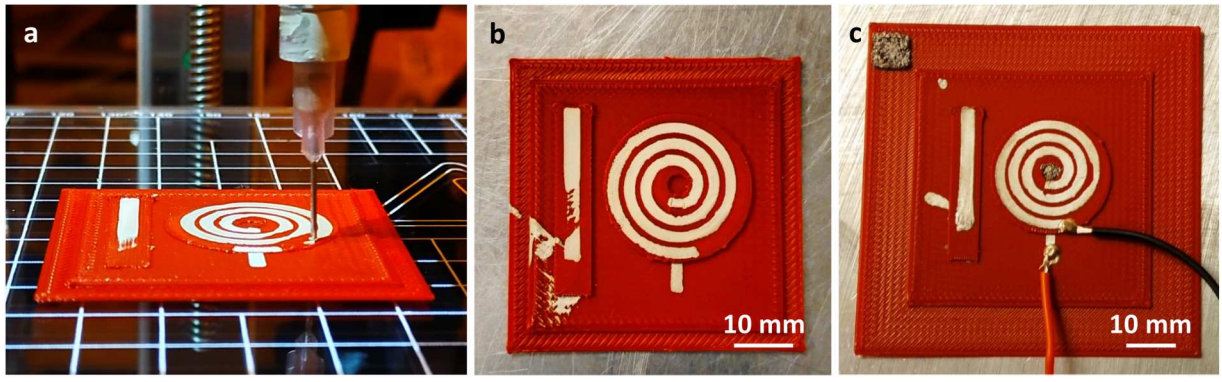


Figure 9. Three-dimensional, silver ink-based stacked solenoids monolithically 3D-printed with the multi-modal, multi-material extrusion system: during printing (a), top view of an air-cored solenoid after printing (b), and top view of a FeSiAl-nylon-cored solenoid after printing (c).

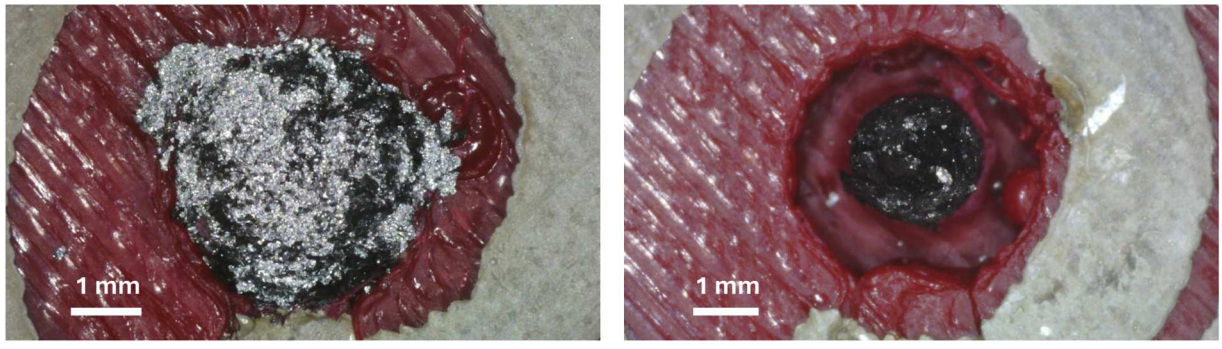


Figure 10. Close-up pictures of monolithically 3D-printed (left) and inserted (right) FeSiAl-nylon soft magnetic cores.

based solenoids. The data show that the cored, silver ink-based solenoids developed in this study significantly outperform those found in the literature, generating magnetic fields almost four times more intense than the highest previously reported [33]. This improvement is attributed to the superior electrical conductivity of silver ink over copper-doped PLA, which enables the application of higher electric currents before incurring excessive heating, and to the inclusion of a high-performance soft magnetic core. Notably, the silver ink-based solenoids developed in this work exhibit a smaller magnetic flux density-versus-current slope compared to the previously reported copper-doped PLA-based solenoids. This is due to the fact that the silver ink-based solenoids incorporate fewer turns: even though they feature the same number of stacked layers (eight), the number of turns per layer is smaller as a result of the brushing technique used to guarantee

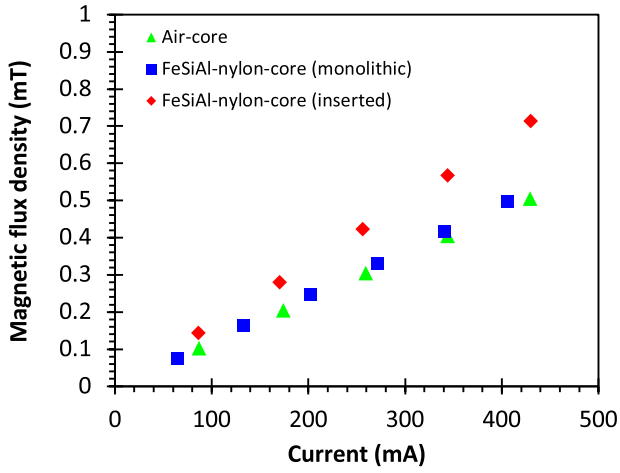


Figure 11. Magnetic flux density vs. current from four-layered, silver ink-based, 3D-printed solenoids with different cores. The error of the magnetic field measurements is 10 mG.

the continuity of the conductive silver ink path, which requires the use of wider traces. However, although silver ink-based solenoids have lower in-plane loop density (i.e. the number of conductive loops per unit area in a layer) than copper-doped PLA-based solenoids, their out-of-plane layer density is higher. This is due to the ability to 3D print very thin (~30 µm) silver ink layers, resulting in more compact devices for the same number of layers. Hence, Figure 12 suggests that, while containing fewer turns and occupying a smaller volume, the silver ink-based solenoids fabricated in this study vastly outperform their previously reported, copper-doped PLA-based counterparts. It is worth noting, however, that the use of silver ink, which requires curing, results in significantly longer print times than when using conductive PLA, which does not require further steps.

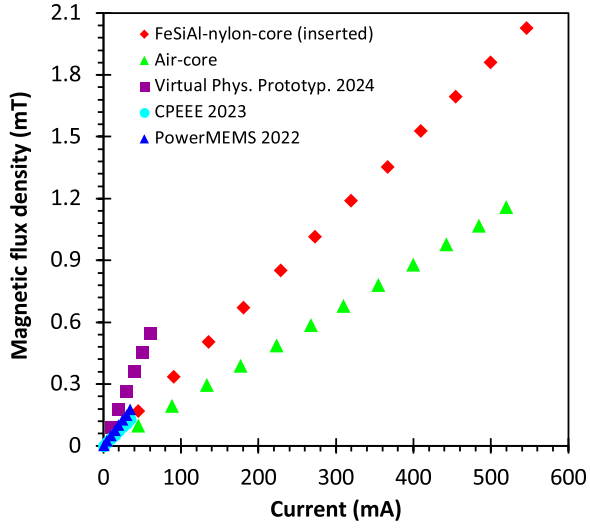


Figure 12. Magnetic flux density vs. current from a silver ink-based solenoid with an inserted FeSiAl-nylon core (red), silver ink-based solenoid with air core (green), copper-PLA-based 3D-printed solenoid with an inserted FeSiAl-nylon core (purple) [33], a monolithically 3D-printed, copper-PLA-based 3D-printed solenoid with an iron-PLA core (cyan) [90], and a copper-PLA-based 3D-printed solenoid with an air core (blue) [89].

3.2. Fully 3D-printed permanent magnets

Four cylindrical permanent magnets of nominal diameter 8 mm and length 2, 4, 6, and 12 mm were 3D printed using the strontium ferrite-doped nylon pellets and the pellet extruder described in Section 2 (Figure 13). The printing settings used when printing the hard magnets are summarised in Table 3. The printing settings employed were optimised via experimental iteration, taking the printable feedstock's manufacturers' recommendations into account.

Nylon 12 often presents adhesion challenges during extrusion 3D printing as it does not stick securely to other polymers or standard print bed surfaces. To alleviate this issue, a thin plate made of FiberFlex 40D was printed beneath the permanent magnets to improve first-layer adhesion using the raft setting of Simplify3D. FiberFlex 40D, a highly elastomeric and adhesive material, provides excellent adhesion across a wide range of substrates. Its use as an intermediary layer significantly improved the interfacial adhesion of the nylon 12-based magnetic composite, enabling reliable and repeatable printing of pellet-based magnetic structures. The rafts were easily detached from the 3D-printed magnets after fabrication. Post-printing, the fabricated components were magnetised by placing them between the poles of a vibrating sample magnetometer (VSM Model 1660, ADE Technologies, Westwood, USA) set to produce a constant magnetic field of 1.5 T.

The 3D-printed magnets were characterised by measuring the remanent magnetic flux density at the intersection of their axis and top surface using a gaussmeter equipped with a transversal probe (HGM0900, MAGSYS magnet systeme GmbH, Dortmud, Germany) (Figure 14).

The external magnetic flux density generated by a uniformly magnetised cylinder along its axis B_z, can be estimated as a function of its magnetisation M, its length L, its radius R, the distance from the surface of the magnet along its axis z, and the permeability of free space μ_0 [91]:

$$B_z(L) = \frac{\mu_0 M}{2}\left[\frac{z+L}{\sqrt{R^2+(z+L)^2}} - \frac{z}{\sqrt{R^2+z^2}}\right] \quad (1)$$

In the case of the 3D-printed magnets,

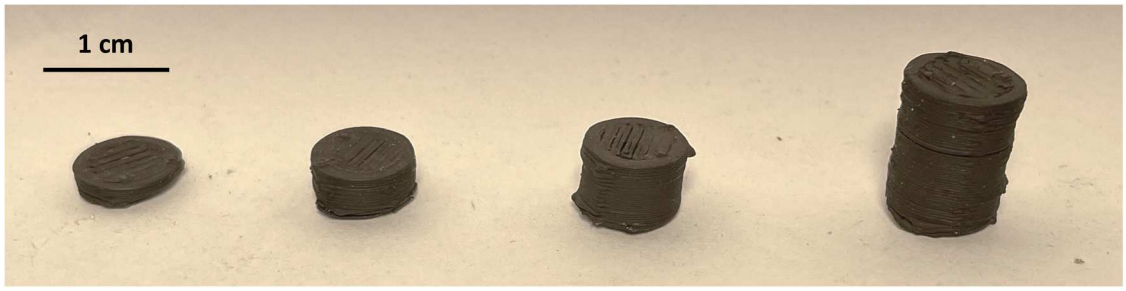


Figure 13. 3D-printed strontium ferrite-doped nylon permanent magnets. The magnets have a diameter of 8.5 mm and heights of (left to right) 2, 4, 6, and 12 mm.

$\mu_0 = 4\pi \times 10^{-7}\ H/m$, $R = 4.25\ mm$ (measured with a caliper), $z = 0.59\ mm$ (distance between the edge of the probe and the measuring plane, according to the probe manufacturer), L is treated as a fitting parameter, and M is considered an unknown. By fitting $B_z(L)$ to the experimental data through nonlinear least-squares numerical optimisation, a reasonable fit to the experimental data was obtained (Figure 14), yielding an estimated magnetisation of $M = 1.25 \times 10^5 A/m$ for the 3D-printed magnets. This value is close to that of the remanent magnetisation obtained independently from the B-H curve of the material in Section 2.2 ($M_r = 1.09\ x10^5\ A/m$), with a deviation of approximately 15%. This discrepancy is attributed to the simplifying assumptions of the analytical field model – most notably uniform magnetisation and ideal cylindrical geometry – which do not fully capture the effects of print-induced layering, porosity, and geometric irregularities. Accordingly, the fit of the experimental data to Equation 1 (Figure 14) exhibits some error, which is consistent with the layered structure of the fabricated magnets resulting from the 3D printing process, rather than perfectly continuous cylinders (Figure 13).

3.3. Fully 3D-printed biaxial bending spring

A two-dimensional spring was designed to behave as the moving component of the 3D-printed motor. The design consists of a central platform set to hold the 3D-printed magnets, connected to a hollow, cylindrical frame by four radial, thin, elastic tethers (Figure 15). The shape of the frame was designed to fit on top of the 3D-printed solenoids described in Section 3.1, while integrating a lateral aperture to facilitate observation. The spring was printed with FiberFlex 40D using the default filament extruder of the E3D ToolChanger described in Section 2, and the printing settings summarised in Table 4.

The spring was designed to be printed upside-down, with the central platform and tethers laying directly on the printer bed, and the structural support standing upwards. This configuration avoids overhanging features and facilitates the monolithic fabrication of the component without requiring the use of supporting material. Additionally, it allows the first layer (i.e. the layer printed directly on the printer bed) to serve as the main feature of the spring. This layer, typically smoother and less prone to geometric mismatch than subsequent layers, forms the tethers of the biaxial spring. To facilitate spring deflection, the tethers were designed to be as thin as possible while guaranteeing structural integrity and adequate deposition of the printable feedstock during 3D printing. It was found experimentally that printing the tethers using a single first layer of 300 µm in thickness yielded a satisfactory compromise between flexibility and integrity, while thinner implementations deformed excessively when removing the part from the printer bed. Figure 16 shows the resulting 3D-printed spring. The biaxial bending springs were not tested as an independent component, but as part of the fully integrated 3D-printed motor (see Section 4).

4. Fabrication and characterisation of fully 3D-printed electric motor

The components described in Section 3 were assembled to build the first reported fully 3D-printed electric motor (Figure 17). One fully 3D-printed electric motor was fabricated and characterised. The 3D-printed permanent magnets of heights 2 and 4 mm were placed on either side of the biaxial bending spring, centred on the platform, magnetically holding each other in place. Once mounted, the magnetic array exhibited a total magnetic remanence of 47 mT, measured at the surface of the largest magnet. The spring was then positioned atop an eight-layers, air-cored, silver ink-based 3D-printed solenoid, with the magnets hovering ~1 mm above the solenoid's centre. When the solenoid is biased at a given voltage, it generates a magnetic field that attracts or repels the 3D-printed magnets, inducing a deflection of the spring and displacing the magnets. The 3D-

Table 3. Summary of key printing parameters used to 3D print strontium ferrite-nylon permanent magnets.

Feature	Tool	Material	Nozzle diameter (mm)	Layer height (µm)	Extrusion multiplier (-)	Extrusion width (mm)	Nozzle Temp. (°C)	Bed Temp. (°C)	Speed (mm/min)	Retraction distance (mm)	Tool change retraction distance (mm)
Permanent magnet	Pellet extruder	Strontium ferrite-nylon	0.4	200	0.67	0.48	280	60	369	2.0	0
Raft	Filament extruder	FiberFlex 40D	0.4	200	1.05	0.48	210	60	369	0.5	1

printed motor was tested both with an air core and with an FeSiAl-doped nylon core inserted at the centre of the solenoid. No significant difference in performance was observed. After assembly of the electric motor, the bottom magnet became slightly tilted with respect to the solenoid (the tilt is visible in Figure 18). The experimental characterisation shows that the slight magnet tilt was not critical to conducting the proof-of-concept demonstration focus of this work.

Two kinds of electrical characterisation of the linear motor were conducted. In the first set of tests, the solenoid was biased at several DC voltages using a TP3005T DC regulated power supply (Tekpower International Co., Ltd., New Taipei City, Taiwan) while recording the position of the magnets (i.e. the centre of the linear motor) using a Moticam S20 camera (Motic, Hong Kong) equipped with Optem FUSION magnification accessories (Qioptiq Photonics, Goettingen, Germany) (Figure 18). Figure 19 depicts the applied bias voltage (from −12 V to 12 V, in increments of 2 V) versus the displacement induced on the magnets, Δz_{DC}, which coincides with the deflection of the biaxial spring at its centre. The DC experimental data show that the deflection of the linear motor has a non-linear dependence on the bias voltage for a large enough voltage value; specifically, the biaxial spring gets stiffer as it is deflected, i.e. larger bias voltage increments are required to further increase the displacement of the motor by a given amount.

The force acting on the 3D-printed biaxial spring is linear with the bias voltage V_{DC}, as the magnetic force is linear with the DC current I_{DC} of the coil. Moreover, a cubic fit closely follows the experimental data (Figure 19), suggesting that the force acting on the spring is proportional to the cube of its deflection:

$$F \propto I_{DC} \propto V_{DC} \propto \Delta z_{DC}^3 \tag{2}$$

This behaviour agrees with the theoretical, quasi-static mechanical performance of a fixed-fixed beam loaded at its centre by a vertical point force when the deflection is large, i.e. the deflection is roughly comparable to the height of the beam. In such a case, the force acting on the beam is proportional to the cube of the deflection of the midpoint of the beam [92], consistently with Equation 2.

In the second set of tests, the motor was excited with 6 V peak-to-peak sinusoidal waveforms with frequencies between 1 Hz and 1 kHz using a WaveStation 2052 arbitrary waveform generator (Teledyne LeCroy, Chestnut Ridge, NY, USA), while recording the oscillation of the spring. Figure 20 shows the evolution of the amplitude of the spring oscillation with the frequency of the applied signal. The data reveal that the resonant

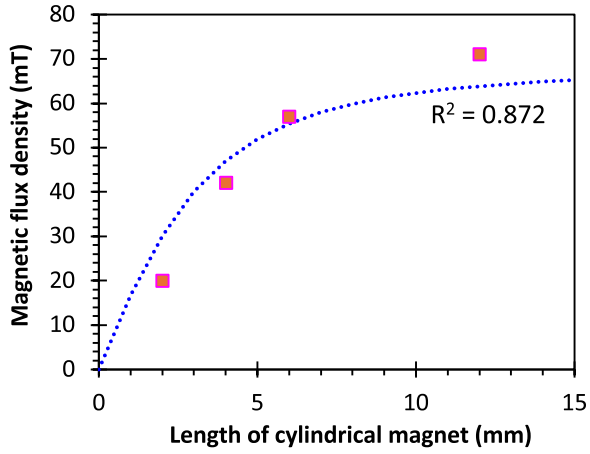


Figure 14. Remanent magnetic flux density generated by the 3D-printed permanent magnets, measured at the intersection between their axis and their top layer, vs. magnet length (makers) and fitted estimate computed according to Equation 1 (dotted line).

frequency of the system is about 41.6 Hz, at which the spring reaches a maximum oscillation amplitude of 318 µm (Figure 21). The estimation of the resonant frequency makes it possible to further characterise the 3D-printed linear motor through the relationship between the resonant frequency and the spring constant:

$$f_{res} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}, \tag{3}$$

where f_{res} is the resonant frequency of the system, k is the spring constant, and m is the mass attached to the spring. The combined mass of the 3D-printed magnets was measured at 1.1 grams, resulting in an estimated spring constant of the biaxial bending spring of $k =$ 75.2 N/m.

5. Discussion

The proof-of-concept electric motor developed in this work is, to the best of our knowledge, the first implementation of a fully 3D-printed motor. It is fabricated using a single AM technology – material extrusion – and all its components can be fabricated using a single multi-modal, multi-material extrusion machine. While the permanent magnets require the use of additional equipment for magnetisation after printing, the rest of the components are used as-printed, resulting in a simple, straightforward manufacturing process. This milestone demonstrates the capability of material extrusion for fabricating electrical machines and paves the way for their monolithic manufacture and integration into fully 3D-printed systems, such as functionalised prosthesis and robots. In this work, component alignment, bonding issues, and mechanical losses were not investigated, as they are not germane to conducting the proof-of-concept demonstration of the electric motor. Nonetheless, these are very important considerations for practical devices (see Sections 5.2 and 5.3).

The next steps towards realising the full potential of material extrusion 3D printing for the fabrication of electrical machines are: (i) integrating the magnetisation of hard magnetic materials into the multi-material extrusion process itself, (ii) demonstrating the fabrication of fully 3D-printed rotary electrical machines, and (iii) achieving the monolithic fabrication of the devices on a single machine, in a single manufacturing step. These goals are briefly discussed right below.

5.1. Magnetising hard magnets during 3D printing

Post-printing magnetisation not only introduces an additional step in the manufacturing process, but also constrains the designs that can be feasibly manufactured. For example, the fully fabricated part must fit between, or sit next to, the magnetising poles; also, adjacent 3D-printed magnets cannot be magnetised independently (e.g. with opposing polarity) with precision. The ability to magnetise hard magnetic materials while they are being 3D printed would

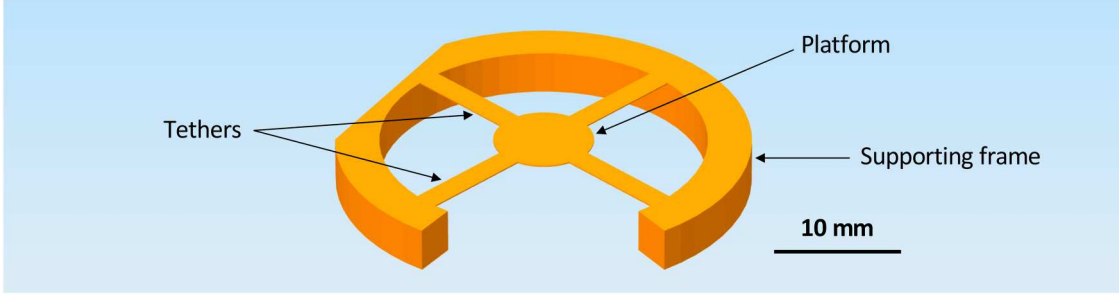


Figure 15. Rendering of two-dimensional spring and supporting frame in Simplify3D.

Table 4. Summary of key printing parameters used to 3D print the biaxial bending spring.

Feature	Tool	Material	Nozzle diameter (mm)	Layer height (µm)	Extrusion multiplier (-)	Extrusion width (mm)	Nozzle Temp. (°C)	Bed Temp (°C)	Speed (mm/min)	Retraction distance (mm)	Tool change retraction distance (mm)
Spring	Filament extruder	FiberFlex 40D	0.4	200 300 (1st)	1.05	0.48	210	60	450	0.5	1

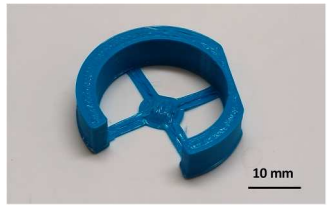


Figure 16. Picture of two-dimensional, bending spring and supporting frame monolithically 3D-printed in FiberFlex 40D. The component is shown in the orientation it is printed in.

eliminate the need for post-printing magnetisation. Recently, magnetic field-assisted 3D printing has been successfully used to induce the self-assembly of microarchitectures, to affect the morphology of 3D-printed devices, and to encode magnetic domains in 3D-printed parts [15]. Remarkably, Kim et al. demonstrated an ink extrusion method for the simultaneous 3D printing and magnetisation of hard magnetic composite ink [93]. The reported technique, which uses an electromagnet attached to a syringe to magnetise the ink as it flows out of the needle, enables the direct fabrication of hard magnets with pre-defined magnetic domains. Furthermore, Kim and colleagues report that a modest magnetic field (50 mT) applied at the needle can yield ~64% of the magnetisation achievable though external magnetisation inside a 2.7 T field, suggesting that, depending on the material used, magnetising while printing can provide competitive results compared to external magnetiation using dedicated equipment. While the proposed technique is limited in terms of the polarisation directions that can be implemented (i.e. to the best of our knowledge, only vertical fields have been demonstrated), it shows great promise for the single-step fabrication of electrical machines including permanent magnets, such as the linear motor demonstrated in this work, and other configurations such as axial flux motors, where all the magnets exhibit magnetic dipole moments parallel to the rotation axis, with alternating signs. In addition to enabling the magnetisation of hard magnets while printing, magnetic field-assisted 3D printing can enhance the directional performance of magnetic cores made of anisotropic soft magnetic materials, such as the one used in this study, by orienting the magnetic filler during the printing operation.

5.2. Fabrication of rotary electrical machines

The presented proof-of-concept of a fully 3D-printed linear motor is a milestone that demonstrates the aptitude of material extrusion for the integral fabrication of electrical machines, and it exhibits immediate translatability for

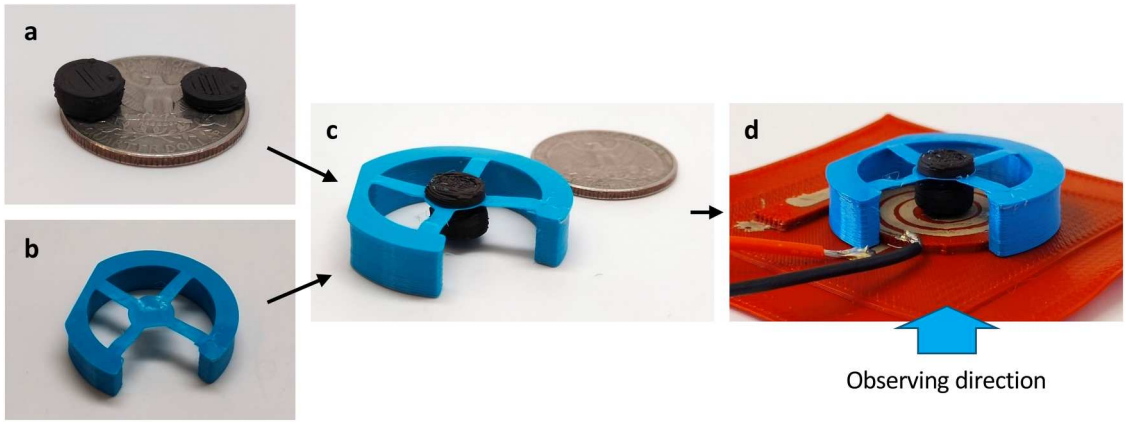


Figure 17. Fully 3D-printed linear motor. Hard magnets (a), biaxial bending spring (b), assembly of spring and magnets (c), assembly of 3D-printed electric motor (d).

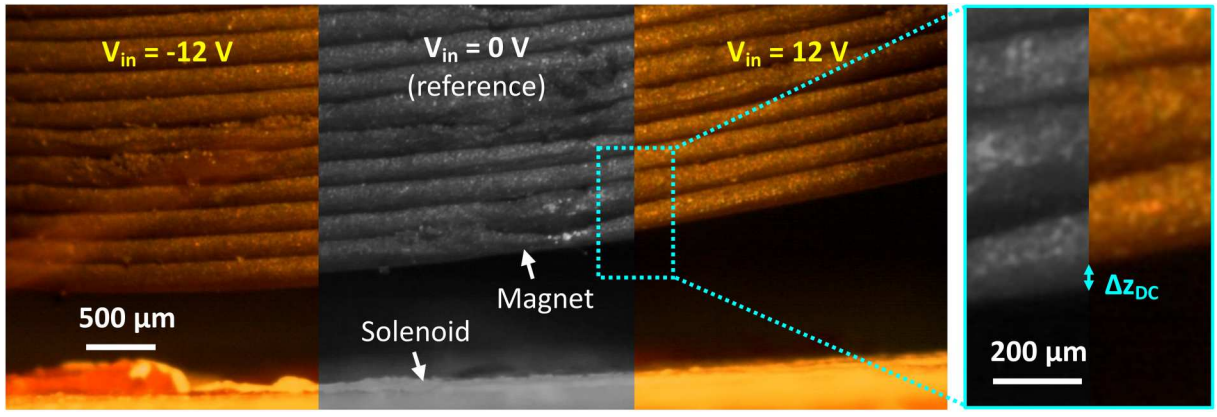


Figure 18. Combined pictures of fully 3D-printed motor without bias (centre, grayscale), biased at – 12 V (left), and biased at 12 V (right), showing detail of the vertical displacement Δz_{DC} of the magnets. Each section of the figure corresponds to one third of a full picture of the gap between the bottom of the bottom permanent magnet and the top of the coil, while the device was biased at different voltages. All three pictures were taken without altering the sample-camera setup to guarantee proper alignment, resulting in the continuous depiction of the top surface of the solenoid across the stitched figure, and variations only in the position of the magnet resulting from the voltage bias.

the fabrication of relevant hardware, such as membrane actuators, pumps, valves, piston mechanisms, and vibratory devices. Using the demonstrated technology, such hardware can be fabricated in situ, with highly customised geometries, and fully integrated into larger 3D-printed devices. However, the full potential of electrical machines is arguably not achievable without the fabrication of rotary devices (e.g. rotary electric motors, generators). The fabrication of rotary electrical machines requires, in addition to the electromagnetic components already demonstrated, the manufacture of mechanical couplings, namely bearings. Works on the AM of bearings have successfully demonstrated fully 3D-printed bearings, including ball bearings [94] and journal bearings [95] fabricated via material extrusion. The aptitude of material extrusion for the monolithic fabrication of bearings was also verified in this work through the single-step manufacture of ball bearings, roller bearings, and planetary gear bearings based on the design by E. Lalish [96] (Figure 22). Even though the designs were not optimised, the printed specimens were found to turn smoothly, suggesting that further investigation on the optimal geometry and lubrication conditions of 3D-printed bearings can yield competitive mechanisms that can be fabricated monolithically and fully integrated in larger, multi-material, 3D-printed devices, such as rotary electrical machines.

5.3. Prospects for the single-step 3D printing of electrical machines

This work has demonstrated the ability of a single multimodal, multi-material extrusion system to produce all the components required for the fabrication of electrical machines. However, the single-step, assembly-free

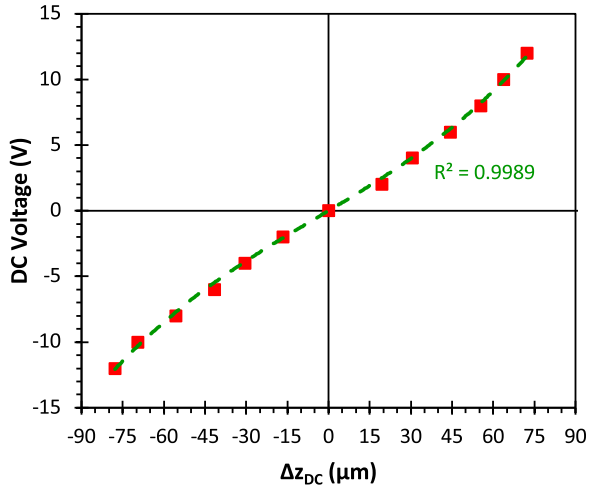


Figure 19. Applied DC bias voltage vs. induced vertical displacement of the magnets placed at the centre of the fully 3D-printed linear motor. The dotted line is a cubic fit of the data.

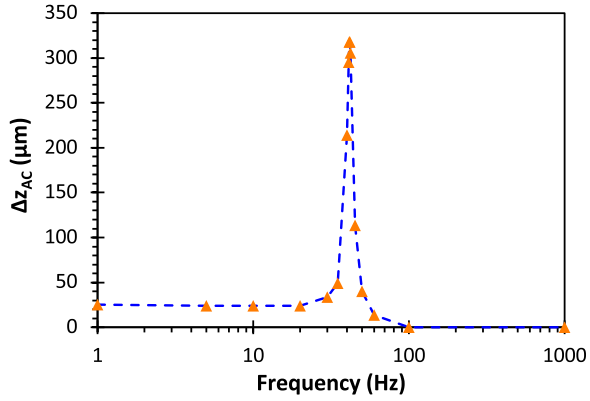


Figure 20. Oscillation amplitude vs. input signal frequency for the fully 3D-printed motor when excited by a sinusoidal input of 6 V peak-to-peak amplitude.

fabrication of electrical machines remains a challenge due to (i) the number of different materials required for their fabrication, (ii) the need to produce mechanically independent parts and overhanging features that are challenging to 3D print, and (iii) the growing complexity of the designs, which drastically increases the probability of critical failures during fabrication. Additionally, the alignment of different components, interfacial adhesion between dissimilar materials, and scalability are critical issues to be addressed for the future production of 3D-printed electrical machines.

In this study, five different materials were processed using a 3D printing system that only accommodates four tools, prompting the need for the sequenced, independent fabrication of components, or the use of multiple 3D printing systems in parallel, plus subsequent assembly. To enable the monolithic fabrication of electrical machines, it is necessary to either increase the number of tools available or decrease the number of tools required. The possibility to simplify the fabrication by decreasing the number of tools required can be addressed by using versatile materials and reusing tools. For example, the same material can be used to fabricate electrically insulating and mechanically active features. In this work, two different materials (PLA and FiberFlex 40D) were used to 3D print electrically insulating layers and compliant features. However, a single material could be used to fabricate both types of structure, provided that its mechanical properties are suited to both tasks. Similarly, a dedicated heater was used to cure conductive silver ink. Alternatively, one of the heated extruders – preferably a pellet extruder, since they work at higher temperatures and are less prone to oozing than filament extruders – could be used as proximity heater for ink curing, at the expense of producing a more localised heating of the sample and hence requiring longer times to sweep the ink patterns. Implementing these strategies, the number of tools required on the 3D printer can be kept low (e.g. four tools: one filament extruder for structural/electrically insulating material, one syringe pump for electrically conductive ink, and two pellet extruders to dispense soft and hard magnetic materials, and to cure ink). However, it is

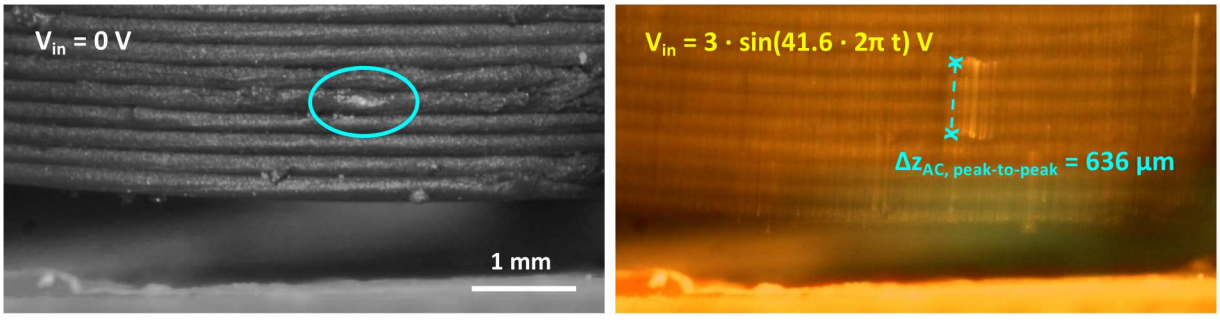


Figure 21. Pictures of the magnets part of the fully 3D-printed motor without bias voltage (left, grayscale) and under a sinusoidal excitation of amplitude 3 V and frequency 41.6 Hz (resonant frequency of the system) (right). Recognisable features were tracked to extract measurements of displacement from the images.



Figure 22. Picture of bearings fabricated in this study to explore the aptitude of material extrusion 3D printing for the fabrication of mechanical couplings, next to a US quarter. From left to right: ball bearing, roller bearing, and planetary gear bearing based on the design by E. Lalish [96].

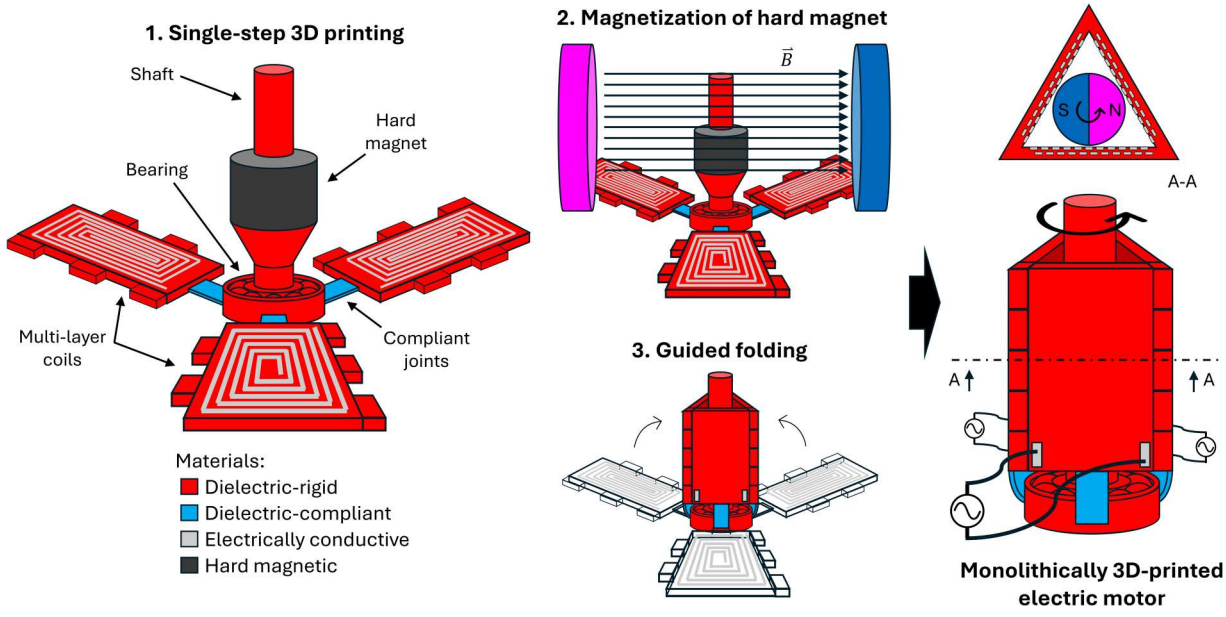


Figure 23. Proposed concept of a rotary electric motor that can be monolithically 3D printed using the hardware and materials presented in this work.

foreseeable that a larger number of tools will result in the ability to fabricate more capable devices and endow the 3D printing system with enhanced versatility. Moreover, the use of additional materials – in particular, soluble materials – can assist in addressing the second challenge (i.e. the fabrication of mechanically independent parts and overhanging features).

The layer-by-layer nature of material extrusion limits its ability to produce overhanging and free-standing features. While it is possible to engineer 3D printing-friendly devices intentionally designed to avoid excessive overhangs (e.g. a rotary electric motor that could be monolithically 3D printed using the hardware and materials presented in this work (Figure 23)), the range of devices that can be feasibly adapted to such constraints is tightly limited. Supports (i.e. vertical, sacrificial structures) 3D printed using the same materials as the main parts can enable the fabrication of overhanging features. However, this is not a suitable approach when dimensional accuracy or surface quality are important, or when the location of the supports complicates their removal (e.g. inside a cavity). Conversely, supports printed using soluble materials are easily removed, enabling the fabrication of intricate, more dimensionally accurate overhangs and free-standing features [97]. By incorporating the use of soluble materials, mechanically independent parts, such as a multiphase rotor and stator for an axial flux motor, could be 3D printed monolithically. This solution reinforces the interest in developing 3D printing systems capable of handling a larger number of distinct materials simultaneously, namely electrically conductive, soft magnetic, hard magnetic, dielectric-rigid, dielectric-compliant, and soluble. It is worth noting that the ability to utilise multiple types

of electrically conductive materials in a single print might also be desirable. For example, highly conductive materials can be used to fabricate coils, more resistive materials can be used to implement compact resistors, and materials with a nonlinear electrical response can enable active electronics [36]. This consideration further increases the number of desirable materials and, therefore, the number of extruders to be incorporated in future multi-material 3D printing systems.

It is also important to highlight that, as the designs grow in complexity, so does the likelihood of unsuccessful 3D printing. This is particularly critical when fabricating electrically conductive features, since a small deviation from the intended design can drastically affect the performance of the finished part (e.g. if several turns of a solenoid are shorted together), or render it useless (e.g. if a circuit is open). Assuming a fixed probability p that a particular layer is printed correctly, the probability that a full device incorporating n layers is printed without issues becomes p^n. If, for example, $p = 0.95$ (i.e. 5% chance of failure in a particular layer) the probability of a device comprising $n = 30$ layers of being printed successfully is only 0.21. This circumstance highlights the need to use highly reliable hardware for 3D printing electrical machines. While the E3D ToolChanger used in this study is a capable system and a powerful research tool because of its customisability, we believe that industry-grade, highly reliable multi-modal, multi-material extrusion systems will be required to successfully realise the 3D printing of electrical machines. As demonstrated in this and other works, multi-material extrusion has the potential to become a truly universal manufacturing technology. A machine capable of reliably processing high-performance conductive materials, soft and hard magnetic composites, rigid, flexible, and soluble materials, could enable the single-step fabrication of electrical machines, as well as electrical machine-enabled complex devices, such as robots, vehicles, and functionalised prostheses – in situ, with high customisability, low-waste, and low-cost.

Envisioning the future of 3D-printed electrical machines beyond the prototyping stage brings forth issues of component alignment, interfacial adhesion, and scalability. In the proof-of-concept demonstration presented here, alignment tolerances between components are relatively forgiving. However, in practical rotary or linear machines, even small misalignments between coils, magnetic cores, and magnets can lead to substantial reductions in torque and efficiency. Single-step (i.e. monolithic) 3D printing inherently resolves alignment concerns, since the assembly step is eliminated and the accuracy of the relative positioning of components becomes determined by the tool displacement resolution of the 3D printer, typically on the order of tens of micrometers. At the same time, monolithic – assembly-free – 3D printing transforms the challenge of adequate component bonding (i.e. holding different parts of the motor together) into an interfacial adhesion matter. Different components serving different functions (e.g. conductive coils, magnetic cores) are expected to be made of different materials, often introducing adhesion challenges [49]. Monolithic devices made from a single base material (e.g. nylon [14], PLA [90]) where different components incorporate distinct fillers to attain or enhance specific properties, do not present this issue provided the filler concentration is not excessive (typically up to about 50% by volume [13]), but their performance is limited. The use of dissimilar materials such as those used in this study enables access to higher performance, but it can aggravate adhesion issues. For example, interfaces such as PLA–nylon or nylon–hard magnetic composites exhibit low affinity and can delaminate under thermal or mechanical stress [98–100]. Addressing interfacial adhesion is essential for reliable, single-step 3D printing of electrical machines, and strategies such as introducing intermediate, compliant materials (e.g. FiberFlex 40D) or gradient-material transitions may play a critical role. These considerations underscore that the transition to monolithic fabrication of electrical machines is not merely a matter of adding tools or materials, but also of engineering the interfaces and positional accuracy that ultimately govern device performance.

Regarding scalability, the presented 3D printing system and printable feedstock can be employed to produce larger electrical machines, up to the build volume of the system (300 mm × 200 mm × 250 mm – typical for desktop 3D printers). The fabrication of larger machines would require the use of larger 3D printing equipment. Higher-power devices are also feasible, provided that thermal management is carefully addressed, for instance, by incorporating a dielectric material with high thermal conductivity to facilitate heat dissipation in the coils [101], or by implementing forced cooling strategies, which may require the creation of cooling channels directly embedded in the device. However, the reliability of the 3D printing process imposes practical limits on scalability, particularly for multi-material, high-precision devices, as previously discussed.

6. Conclusion

In this study, a multi-modal, multi-material extrusion system capable of processing feedstock in the form of filament, pellets, and ink was developed, enabling the combined use of high-performance functional materials.

The 3D printing system, derived from a commercial desktop 3D printer, was used to fabricate the totality of the critical components that build up electrical machines, namely electrically conductive solenoids, soft magnetic cores, hard magnets, and mechanical couplings. With minimal assembly, the first fully 3D-printed motor was created in this work, showcasing the potential of the technology for the monolithic manufacture of complex electromagnetic hardware. In the process, fully 3D-printed solenoids capable of generating magnetic fields of 2 mT – almost four times more intense than those previously reported – were demonstrated. The proof-of-concept motor, which can be described as a solenoid-magnet linear actuator, is the first electrical machine whose components have been fully 3D printed using a single technology – material extrusion – with the magnetisation of the hard magnets being the only postprocessing step required. This milestone attests to the potential of material extrusion to become an almost universal manufacturing technology with the ability to leverage high-performance materials and combine them to create highly functional devices. Further development of the technology can enable the monolithic, single-step fabrication of electrical machines, eliminating the need for off-the-shelf components, post-processing, and complex assembly. Such advancements would not only streamline the production of plug-and-play electrical machines but also facilitate their seamless integration into more complex 3D-printed devices, such as functionalised prosthesis and robots, while leveraging high customisability and in-situ, low-cost, low-waste manufacturing.

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Author contributions

CRediT: **Jorge Cañada:** Conceptualization, Data curation, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Zoey Bigelow:** Investigation, Validation, Visualization, Writing – review & editing; **Luis Fernando Velásquez-García:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability

The data that support the findings of this study are available in the repository https://doi.org/10.7910/DVN/HAWSTW.

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References

[1] Quanjin M, Rejab MRM, Idris MS, et al. Recent 3D and 4D intelligent printing technologies: a comparative review and future perspective. Procedia Comput Sci. 2020;167:1210–1219. doi:10.1016/j.procs.2020.03.434
[2] Eckhoff CC, Lubinsky NK, Metzler LJ, et al. Low-cost, compact quadrupole mass filters with unity mass resolution via ceramic resin Vat photopolymerization. Advanced Science. 2024;11(9):2307665. doi:10.1002/advs.202307665
[3] Buchner TJK, Rogler S, Weirich S, et al. Vision-controlled jetting for composite systems and robots. Nature. 2023;623(7987):522–530. doi:10.1038/s41586-023-06684-3
[4] Li F, Macdonald NP, Guijt RM, et al. Increasing the functionalities of 3D printed microchemical devices by single material, multimaterial, and print-pause-print 3D printing. Lab Chip. 2019;19(1):35–49. doi:10.1039/c8lc00826d
[5] Máximo DM, Velásquez-García LF. Additively manufactured electrohydrodynamic ionic liquid pure-ion sources for nanosatellite propulsion. Additive Manufacturing. 2020;36:101719. doi:10.1016/j.addma.2020.101719
[6] Izquierdo-Reyes J, Bigelow Z, Lubinsky NK, et al. Compact retarding potential analyzers enabled by glass-ceramic vat polymerization for CubeSat and laboratory plasma diagnostics. Addit Manuf. 2022;58:103034. doi:10.1016/j.addma.2022.103034
[7] Kim H, Velásquez-García LF. High-impulse, modular, 3D-printed CubeSat electrospray thrusters throttleable via pressure and voltage control. Advanced Science. 2025;12(13):2413706. doi:10.1002/advs.202413706

[8] Berman B. 3-D printing: The new industrial revolution. Bus Horiz. 2012;55(2):155–162. ISSN 0007-6813. doi:10.1016/j.bushor.2011.11.003
[9] Rayna T, Striukova L. From rapid prototyping to home fabrication: How 3D printing is changing business model innovation. Technol Forecast Soc Change. 2016;102:214–224. ISSN 0040-1625. doi:10.1016/j.techfore.2015.07.023
[10] Lee H-J, Cañada J, Velásquez-García LF. Compact peristaltic vacuum pumps via multi-material extrusion. Addit Manuf. 2023;68:103511. doi:10.1016/j.addma.2023.103511
[11] Hubbard JD, et al. Fully 3D-printed soft robots with integrated fluidic circuitry. Sci Adv. 2021;7(29):eabe5257. doi:10.1126/sciadv.abe5257
[12] Taylor AP, Velásquez-García LF. Miniaturized diaphragm vacuum pump by multi-material additive manufacturing. J Microelectromech Syst. 2017;26(6):1316–1326. doi:10.1109/JMEMS.2017.2743020
[13] Taylor AP, Vélez Cuervo C, Arnold D, et al. Fully 3D-printed, monolithic, mini magnetic actuators for low-cost, compact systems. J Microelectromech Syst. 2019;28(3):481–493. doi:10.1109/JMEMS.2019.2910215
[14] Taylor AP, Reyes JI, Velásquez-García LF. Compact, magnetically actuated, additively manufactured pumps for liquids and gases. J Phys D: Appl Phys. 2020;53(35):355002. doi:10.1088/1361-6463/ab8de8
[15] Zhang C, Li X, Jiang L, et al. 3D printing of functional magnetic materials: from design to applications. Adv Funct Mater. 2021;31(34):2102777. doi:10.1002/adfm.202102777
[16] Shao G, Hai R, Sun C. 3D printing customized optical lens in minutes. Adv Opt Mater. 2020;8(4):1901646. doi:10.1002/adom.201901646
[17] Gao H, et al. 3D printed optics and photonics: processes, materials and applications. Mater Today. 2023;69:107–132. doi:10.1016/j.mattod.2023.06.019
[18] Espera AH, Dizon JRC, Chen Q, et al. 3D-printing and advanced manufacturing for electronics. Progress in Addit Manuf. 2019;4(3):245–267. doi:10.1007/s40964-019-00077-7
[19] Sun Z, Velásquez-García LF. Monolithic FFF-printed, biocompatible, biodegradable, dielectric-conductive microsystems. J Microelectromech Syst. 2017;26(6):1356–1370. doi:10.1109/JMEMS.2017.2746627
[20] Flowers PF, Reyes C, Ye S, et al. 3D printing electronic components and circuits with conductive thermoplastic filament. Addit Manuf. 2017;18:156–163. doi:10.1016/j.addma.2017.10.002
[21] Wu F, El-Refaie AM. "Towards fully additively-manufactured permanent magnet synchronous machines: opportunities and challenges," 2019 *IEEE International Electric Machines & Drives Conference*, San Diego, CA, USA, pp. 2225-2232, 2019. doi:10.1109/IEMDC.2019.8785210
[22] Rao CH, Avinash K, Varaprasad BKSVL, et al. A review on printed electronics with digital 3D printing: fabrication techniques, materials, challenges and future opportunities. J Electron Mater. 2022;51(6):2747–2765. doi:10.1007/s11664-022-09579-7
[23] Tan HW, Choong YYC, Kuo CN, et al. 3D printed electronics: processes, materials and future trends. Prog Mater Sci. 2022;127:100945. doi:10.1016/j.pmatsci.2022.100945
[24] Saengchairat N, Tran T, Chua CK. A review: additive manufacturing for active electronic components. Virtual Phys Prototyp. 2017;12(1):31–46. doi:10.1080/17452759.2016.1253181
[25] Ding C, et al. "Magnetic paste as feedstock for additive manufacturing of power magnetics," 2018 *IEEE Applied Power Electronics Conference and Exposition*, San Antonio, TX, USA, pp. 615-618, 2018. doi:10.1109/APEC.2018.8341075
[26] Liu L, Ngo KDT, Lu G-Q. Effects of Co_3O_4 addition on magnetic properties of NiCuZn ferrite feedstock for 3-D-printing power magnetic components. IEEE Trans Magn. 2020;56(12):1–7. doi:10.1109/TMAG.2020.3030760
[27] Bollig LM, et al. 3D printed magnetic polymer composite transformers. J Magn Magn Mater. 2017;442:97–101. doi:10.1016/j.jmmm.2017.06.070
[28] Bowen D, Basu D. Exploration of the 3-D printing of transformer structures with embedded ferrite cores. IEEE Trans Magn. 2019;55(2):1–5. doi:10.1109/TMAG.2018.2875811
[29] Cañada J, Nagle SF, Vidal N, et al. 3D-printed soft magnetic cores for compact electromechanical devices via material extrusion. 2024 *IEEE 23rd International Conference on Micro and Miniature Power Systems, Self-Powered Sensors and Energy Autonomous Devices (PowerMEMS)*, Tonsberg, Norway. 2024: 255–258. doi:10.1109/PowerMEMS63147.2024.10814238
[30] Simpson N, Yiannakou G, Felton H, et al. Direct thermal management of windings enabled by additive manufacturing. IEEE Trans Ind Appl. 2023;59(2):1319–1327. doi:10.1109/TIA.2022.3209171
[31] Hou T, et al. Design of 3D wireless power transfer system based on 3D printed electronics. IEEE Access. 2019;7:94793–94805. doi:10.1109/ACCESS.2019.2928948
[32] Mettes S, Kohls N, Allen KW, et al. Characterization of a 3D printed electric solenoid with multi-layer silver nanoparticle traces. IFAC-PapersOnLine. 2022;55(37):139–144. ISSN 2405-8963. doi:10.1016/j.ifacol.2022.11.174
[33] Cañada J, Kim H, Velásquez-García LF. Three-dimensional, soft magnetic-cored solenoids via multi-material extrusion. Virtual Phys Prototyp. 2024;19(1):e2310046. doi:10.1080/17452759.2024.2310046
[34] Taylor AP, et al. Compact, magnetically actuated, additively manufactured pumps for liquids and gases. J Phys D: Appl Phys. 2020;53(35):355002. doi:10.1088/1361-6463/ab8de8
[35] Lee YJ, Lee KH, Lee CH. Friction performance of 3D printed ball bearing: feasibility study. Results Phys. 2018;10:721–726. doi:10.1016/j.rinp.2018.07.011
[36] Cañada J, Velásquez-García LF. Semiconductor-free, monolithically 3D-printed logic gates and resettable fuses. Virtual Phys Prototyp. 2024;19(1):e2404157. doi:10.1080/17452759.2024.2404157
[37] Lorenz F, Rudolph J, Wemer R. "Design of 3D printed high performance windings for switched reluctance machines," 2018 *XIII International Conference on Electrical Machines (ICEM)*, Alexandroupoli, Greece, 2018, pp. 2451-2457. doi:10.1109/ICELMACH.2018.8506845
[38] Mettes S, Schwalbe JF, Allen KW, et al. Fully 3D-printed multilayer axial flux stator for rotary electromagnetic

motors. IEEE/ASME Trans Mechatron. 2024;29(4):2903–2910. doi:10.1109/TMECH.2024.3400811
[39] Ellery A, Elaskri A, Paranthaman MP, et al. Experiments in 3D printing electric motors. Machines. 2025;13(8):679. doi:10.3390/machines13080679
[40] Goh GL, Zhang H, Chong TH, et al. 3D printing of multilayered and multimaterial electronics: a review. Adv Electron Mater. 2021;7(10). doi:10.1002/aelm.202100445
[41] Li Y, Wu Z, Chen Y, et al. Multi-material embedded 3D printing for one-step manufacturing of multifunctional components in soft robotics. Addit Manuf. 2024;85:104178. doi:10.1016/j.addma.2024.104178
[42] Carrillo CS, Sanchez M. Design and 3D printing of four multimaterial mechanical metamaterial using PolyJet technology and digital materials for impact injury prevention. *in 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society*. 2021: 4916–4919. doi:10.1109/EMBC46164.2021.9630675
[43] Yang F, Ding W, Jia J, et al. 4D printing for bone implant: progress, advantages and challenges. Prog Mater Sci. 2026;157:101591. doi:10.1016/j.pmatsci.2025.101591
[44] Guo W, Li P, Wei Y, et al. Ionic substitution through bredigite doping for microstructure and performance adjustment in DLP 3D-printed TPMS porous HA bone scaffolds. Virtual Phys Prototyp. 2024;19(1):e2423840. doi:10.1080/17452759.2024.2423840
[45] Guo W, Feng S, Li B, et al. Niobium carbide-functionalized polylactic acid bone scaffold with near-infrared light-mediated shape memory and photothermal therapy for complex bone defects. Int J Biol Macromol. 2025;322(1):146578. ISSN 0141-8130. doi:10.1016/j.ijbiomac.2025.146578
[46] Guo W, Wei Y, Xu C, et al. Functionalized degradable soybean oil-based biomimetic porous scaffolds for complex bone defects: Vat photopolymerization additive manufacturing, photothermal-mediated shape memory and tumor thermotherapy. Compos Sci Technol. 2026;273:111412. ISSN 0266-3538. doi:10.1016/j.compscitech.2025.111412
[47] Guo W, Tang X, Li B, et al. Multi-element CuFeSe2 nanoparticle-functionalized 3D-printed polylactic acid scaffolds with photothermal shape memory and therapy for complex bone defects. Ceram Int. 2025. ISSN 0272-8842. doi:10.1016/j.ceramint.2025.10.016
[48] Ligon SC, Liska R, Stampfl J, et al. Polymers for 3D printing and customized additive manufacturing. Chem Rev. 2017;117(15):10212–10290. doi:10.1021/acs.chemrev.7b00074
[49] Lopes LR, Silva AF, Carneiro OS. Multi-material 3D printing: The relevance of materials affinity on the boundary interface performance. Addit Manuf. 2018;23:45–52. doi:10.1016/j.addma.2018.06.027
[50] Gibson I, Rosen D, Stucker B, et al. *Additive manufacturing technologies*. Cham: Springer; 2021. doi:10.1007/978-3-030-56127-7
[51] Loh GH, Pei E, Gonzalez-Gutierrez J, et al. An overview of material extrusion troubleshooting. Appl Sci. 2020;10(14):4776. doi:10.3390/app10144776
[52] Saadi MASR, Maguire A, Pottackal NT, et al. Direct ink writing: a 3D printing technology for diverse materials. Adv Mater. 2022;34(28):2108855. doi:10.1002/adma.202108855
[53] Shaik YP, Schuster J, Shaik A. A scientific review on various pellet extruders used in 3D printing FDM processes. OALib. 2021;08(08):1–19. doi:10.4236/oalib.1107698
[54] Goh G, et al. A 3D printing-enabled artificially innervated smart soft gripper with variable joint stiffness. Advanced Materials Technologies. 2023;8(24):2301426. doi:10.1002/admt.202301426
[55] Rocca M, Fragasso A, Liu W, et al. Embedded multimaterial extrusion bioprinting. SLAS Technology. 2018;23(2):154–163. ISSN 2472-6303. doi:10.1177/2472630317742071
[56] Hoque ME, Chuan YL, Pashby I. Extrusion based rapid prototyping technique: An advanced platform for tissue engineering scaffold fabrication. Biopolymers. 2012;97(2):83–93. doi:10.1002/bip.21701
[57] Górski F, Wichniarek R, Kuczko W, et al. Study on properties of automatically designed 3D-printed customized prosthetic sockets. Materials (Basel). 2021;14(18):5240. doi:10.3390/ma14185240
[58] Zander NE. (2019). Recycled polymer feedstocks for material extrusion additive manufacturing. In *Polymer-based additive manufacturing: Recent developments*; American Chemical Society: Washington, DC, USA; pp. 37–51. ISBN 978-0-8412-3426-0. doi:10.1021/bk-2019-1315.ch003
[59] Tümer EH, Erbi HY. Extrusion-based 3D printing applications of PLA composites: a review. Coatings. 2021;11(4):390. doi:10.3390/coatings11040390
[60] Jiang Z, Diggle B, Tan ML, et al. Extrusion 3D printing of polymeric materials with advanced properties. Adv Sci. 2020;7(17):2001379. doi:10.1002/advs.202001379
[61] Bourell D, et al. Materials for additive manufacturing. CIRP Ann. 2017;66(2):659–681. ISSN 0007-8506. doi:10.1016/j.cirp.2017.05.009
[62] Ding C, et al. "Magnetic paste as feedstock for additive manufacturing of power magnetics," 2018 *IEEE applied power electronics conference and exposition*, San Antonio, TX, USA, pp. 615-618, 2018. doi:10.1109/APEC.2018.8341075
[63] Martin V, et al. Low cost 3D printing of metals using filled polymer pellets. HardwareX. 2022;11:e00292. doi:10.1016/j.ohx.2022.e00292
[64] Engberg A, Stelzl C, Eriksson O, et al. An open source extrusion bioprinter based on the E3D motion system and tool changer to enable FRESH and multimaterial bioprinting. Sci Rep. 2021;11(1):21547. doi:10.1038/s41598-021-00931-1
[65] Hasni U, Green R, Filippas AV, et al. One-step 3D-printing process for microwave patch antenna via conductive and dielectric filaments. Microw Opt Technol Lett. 2019;61(3):734–740. doi:10.1002/mop.31607
[66] Wiklund J, et al. A review on printed electronics: fabrication methods, inks, substrates, applications and environmental impacts. J Manuf Mater Process. 2021;5(3):89. doi:10.3390/jmmp5030089
[67] Cano-Raya C, Denchev ZZ, Cruz SF, et al. Chemistry of solid metal-based inks and pastes for printed electronics – A review. Appl Mater Today. 2019;15:416–430. ISSN 2352-9407. doi:10.1016/j.apmt.2019.02.012
[68] Jiang P, Ji Z, Zhang X, et al. Recent advances in direct ink writing of electronic components and functional devices. Progre Addit Manuf. 2018;3(1-2):65–86. doi:10.1007/s40964-017-0035-x

[69] Ho M, Ramirez AB, Akbarnia N, et al. Direct ink writing of conductive hydrogels. Adv Funct Mater. 2025;35(22):2415507. doi:10.1002/adfm.202415507
[70] Copinet A, Bertrand C, Govindin S, et al. Effects of ultraviolet light (315 nm), temperature and relative humidity on the degradation of polylactic acid plastic films. Chemosphere. 2004;55(5):763–773. ISSN 0045-6535. doi:10.1016/j.chemosphere.2003.11.038
[71] Wang FH, et al. 3D printing of soft magnetic materials: from printing to applications. J Alloys Compd. 2024;990:174486. doi:10.1016/j.jallcom.2024.174486
[72] Bollig LM, Patton MV, Mowry GS, et al. Effects of 3-D printed structural characteristics on magnetic properties. IEEE Trans Magn. 2017;53(11):1–6. Art no. 2300806. doi:10.1109/TMAG.2017.2698034
[73] Patton MV, et al. Manipulating magnetic anisotropy in fused filament fabricated parts via macroscopic shape, mesoscopic infill orientation, and infill percentage. Addit Manuf. 2019;27:482–488. doi:10.1016/j.addma.2019.03.026
[74] Henderson L, et al. Altering magnetic properties of iron filament PLA using magnetic field assisted additive manufacturing (MFAAM). J Magn Magn Mater. 2021;538:168320. ISSN 0304-8853. doi:10.1016/j.jmmm.2021.168320
[75] Díaz-García A, et al. Functional, thermal and rheological properties of polymer-based magnetic composite filaments for additive manufacturing. Mater Des. 2022;219:110806. doi:10.1016/j.matdes.2022.110806
[76] Khatri B, et al. A 3D-printable polymer-metal soft-magnetic functional composite—development and characterization. Materials (Basel). 2018;11(2):189. doi:10.3390/ma11020189
[77] Trnka N, Rudolph J, Werner R. "Magnetic properties of ferromagnetic materials produced by 3D multi-material printing," *IEEE 29th international symposium on industrial electronics*, delft, Netherlands, 326-331, 2020. doi:10.1109/ISIE45063.2020.9152231
[78] Liu L, Ngo KDT, Lu G-Q. Effects of Co_3O_4 addition on magnetic properties of NiCuZn ferrite feedstock for 3-D-printing power magnetic components. IEEE Trans Magn. 2020;56(12):1–7. doi:10.1109/TMAG.2020.3030760
[79] Huber C, Abert C, Bruckner F, et al. 3D print of polymer bonded rare-earth magnets, and 3D magnetic field scanning with an end-user 3D printer. Appl Phys Lett. 2016;109(16):162401. doi:10.1063/1.4964856
[80] Huber C, Abert C, Bruckner F, et al. 3D printing of polymer-bonded rare-earth magnets with a variable magnetic compound fraction for a predefined stray field. Sci Rep. 2017;7:9419. doi:10.1038/s41598-017-09864-0
[81] Pigliaru L, Rinaldi M, Ciccacci L, et al. 3D printing of high performance polymer-bonded PEEK-NdFeB magnetic composite materials. Func Compos Mater. 2020;1(1):4. doi:10.1186/s42252-020-00006-w
[82] Peng E, Wei X, Herng TS, et al. Ferrite-based soft and hard magnetic structures by extrusion free-forming. RSC Adv. 2017;7(43):27128–27138. doi:10.1039/C7RA03251J
[83] Li L, Tirado A, Nlebedim I, et al. Big area additive manufacturing of high performance bonded NdFeB magnets. Sci Rep. 2016;6(1):36212. doi:10.1038/srep36212
[84] Li L, Jones K, Sales B, et al. Fabrication of highly dense isotropic Nd-Fe-B nylon bonded magnets via extrusion-based additive manufacturing. Addit Manuf. 2018;21:495–500. ISSN 2214-8604. doi:10.1016/j.addma.2018.04.001
[85] Paranthaman MP, Yildirim V, Lamichhane TN, et al. Additive manufacturing of isotropic NdFeB PPS bonded permanent magnets. Materials (Basel). 2020;13(15):3319. doi:10.3390/ma13153319
[86] Ntagios M, Nassar H, Pullanchiyodan A, et al. Robotic hands with intrinsic tactile sensing via 3D printed soft pressure sensors. Adv Intell Syst. 2020;2(6):1900080. doi:10.1002/aisy.201900080
[87] Georgopoulu A, Sebastian T, Clemens F. Thermoplastic elastomer composite filaments for strain sensing applications extruded with a fused deposition modelling 3D printer. Flex Print Electron. 2020;5(3):035002. doi:10.1088/2058-8585/ab9a22
[88] Cao X, Xuan S, Gao Y, et al. 3D printing ultraflexible magnetic actuators via screw extrusion method. Adv Sci. 2022;9(16):2200898. doi:10.1002/advs.202200898
[89] Cañada J, Velásquez-García LF. "Fully 3D-printed solenoids for compact systems," *21st international conference on micro and nanotechnology for power generation and energy conversion applications (PowerMEMS)*, Salt Lake City, UT, USA, 2022, pp. 150–153. doi:10.1109/PowerMEMS56853.2022.10007551
[90] Cañada J, Velásquez-García LF. "Monolithically 3D-printed, miniature solenoids with soft magnetic core for compact systems," 13th international conference on power, energy and electrical engineering (CPEEE), Tokyo, Japan, 2023, pp. 116–120. doi:10.1109/CPEEE56777.2023.10217558
[91] Haus HA, Melcher JR. *Electromagnetic fields and energy*. Englewood Cliffs (NJ): Prentice-Hall; 1989.
[92] Senturia S. *Microsystems design*. New York, NY, USA: Springer; 2001. doi:10.1007/b117574
[93] Kim Y, Yuk H, Zhao R, et al. Printing ferromagnetic domains for untethered fast-transforming soft materials. Nature. 2018;558:274–279. doi:10.1038/s41586-018-0185-0
[94] Harikrishnan U, Soundarapandian S. Fused deposition modelling based printing of full complement bearings. Procedia Manuf. 2018;26:818–825. ISSN 2351-9789. doi:10.1016/j.promfg.2018.07.102
[95] Lee Y-J, Lee K-H, Lee C-H. Self-lubricating and friction performance of a three-dimensional-printed journal bearing. J Tribol. 2018;140(5):054501. doi:10.1115/1.4039995
[96] Lalish E. "Gear bearing," Thingiverse, 2013, [cited 2025 October]. Available From: https://www.thingiverse.com/thing:53451
[97] Park SJ, Lee JE, Park JH, et al. Enhanced solubility of the support in an FDM-based 3D printed structure using hydrogen peroxide under ultrasonication. Adv Mater Sci Eng. 2018;2018(1):3018761. doi:10.1155/2018/3018761
[98] Tamburrino F, Graziosi S, Bordegoni M. The influence of slicing parameters on the multi-material adhesion mechanisms of FDM printed parts: an exploratory study. Virtual Phys Prototyp. 2019;14(4):316–332. doi:10.1080/17452759.2019.1607758

[99] Goh GL, Lee S, Cheng SH, et al. Enhancing interlaminar adhesion in multi-material 3D printing: A study of conductive PLA and TPU interfaces through fused filament fabrication. Mater Sci Addit Manuf. 2024;3(1):2672. doi:10.36922/msam.2672

[100] Penter L, Maier J, Kauschinger B, et al. 3D printing technology for low cost manufacturing of hybrid prototypes from multi material composites. *in Congress of the German academic association for production technology*. 2021: 396–405. doi:10.1007/978-3-662-62138-7_40

[101] Mettes S, Bates J, Allen KW, et al. A fully 3D printed, multi-material, and high operating temperature electromagnetic actuator. 2023 *IEEE/ASME international conference on advanced intelligent mechatronics (AIM)*. 2023: 517–524. doi:10.1109/AIM46323.2023.10196155